

The Successful Mathematical Derivation of the Forward-Reverse Reversible Bidirectional Closure of the Riemann Hypothesis

(From the Successful Physical and Mathematical Contradiction Self-Detonation of Structured Reverse
Derivation to the Forward Derivation and Instantaneous Establishment of the Riemann Hypothesis)

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Abstract: Regarding the Riemann Hypothesis, traditional methods rely on zero-by-zero verification or asymptotic estimates in finite regions, whose essence is to implement step-by-step approximation to the infinite whole by finite steps. No matter how much the computational range is expanded, the extrapolation of finite data cannot logically constitute true coverage of the infinite zero set of the Riemann Hypothesis. Methodologically speaking, the limit of traditional approximation paths is “describing the infinite,” while true “covering the infinite” cannot be realized at the definitional level of the Riemann Hypothesis. That is, achieving infinite coverage of the Riemann Hypothesis is a prerequisite for solving the problem, which exceeds the boundary of capability of traditional methodology. In contrast, this paper adopts a strategy of structured reverse derivation, independent of the extrapolation or superposition of local data. Theorem 9.3A reduces the zero problem of doubly reflected entire functions to the zero distribution of the real entire function $g(w)$; the Mother Equation, as a dual-channel unitary system, imposes global structural constraints on arbitrary zero configurations through its unitarity axiom; once the configuration of non-real zeros is incorporated into this framework, it immediately forms a self-detonation of contradictions with the axiom at the definitional level. This mechanism does not rely on zero-by-zero enumeration, but closes the living space of infinite non-real zeros in one stroke at the structural level, achieving global coverage of the infinite zero set of the Riemann Hypothesis. The repulsion of the infinite set of non-real zeros renders the establishment of the Riemann Hypothesis a logical factual consequence of definitional self-consistency, prompting the success of forward derivation and the formation of bidirectional closure.

Keywords: Riemann Hypothesis; Covering the Infinite; Theorem 9.3A; Structured Reverse Derivation; Contradiction Self-Detonation; Forward Derivation; Bidirectional Closure

Methodological Declaration

“Some of the most profound orders in nature are not inscribed in local details, but woven into global structures.” This is precisely why structuring can solve certain problems that formalized local segmentation cannot. Not all problems in nature can be segmented; once segmented, the whole is destroyed. Just as a human being cannot be segmented or partitioned, once broken apart, its integrity is not only destroyed, but can never be restored or reconstructed.

The essence of the Riemann Hypothesis is profoundly abstract. Its common formulation, “All non-trivial zeros of $\zeta(s)$ lie on the line $\text{Re}(s) = 1/2$,” appears simple and highly concrete, yet it is merely a phantom of the true Riemann Hypothesis. To proceed from this formulation is to fall immediately into a double trap: first, the trap of the misplaced object $\zeta(s)$, since $\zeta(s)$ itself lacks the structural integrity required to address the original problem, as

detailed in the preliminary legitimacy argument; second, the trap of the parameter, since $1/2$ is not, as commonly assumed, an external parameter to be verified, but rather a geometric position intrinsically bound to the variable substitution, reducing what is intrinsically a structural problem of global self-consistency for a doubly reflected entire function to an asymptotic problem about point-set distribution on the complex plane.

The true dilemma and difficulty of the Riemann Hypothesis lie in the following: if one adheres to the highly abstract nature of the 1859 German original, namely entire function, even symmetry, and real-conjugate symmetry, one cannot establish a concretized goal of **“alle Wurzeln reell”** that simultaneously satisfies all these requirements; work would be over before it even begins. If, on the other hand, one downgrades from the original $\xi(t)$ to $\zeta(s)$, one may obtain a concretizable objective, but the two are equivalent only within set theory; in every other branch of mathematics they are not identical, and one immediately falls into the trap. That is, the true difficulty of the Riemann Hypothesis in its German original far exceeds the common formulation based on $\zeta(s)$ that has resulted from Stieltjes's downgrade and simplifying compromise since 1885^[3].

For the highly abstract problem of the Riemann Hypothesis in its German original, the Mother Equation is the only choice after 167 years of no way out, completing the concretization objective for the first time. However, the Mother Equation is not designed as a specific customization for the Riemann Hypothesis as a particular problem; had it truly been so customized, its structure would never have taken such a form.

Special Emphasis 1

The reason why the Mother Equation is first defined as the Whole 1 is that formalized mathematics can "describe the infinite" (through limits, induction, universal quantifiers $\forall$, etc.), yet cannot "truly cover the infinite" at the definitional level of the Riemann Hypothesis. The core contradiction is that the Riemann Hypothesis concerns the global distribution of an infinite set of zeros, while all tools of formalized proof are essentially finite. Furthermore, if one follows the traditional path of approximating the infinite by the finite, the proposition of the Riemann Hypothesis, which requires infinite coverage, can never be traversed. The failures of various formalized attempts to approximate the infinite by the finite over the past 167 years have already demonstrated by fact that this path is fundamentally impassable. The value of first defining the Mother Equation as the Whole 1 lies precisely in no longer attempting to derive through finite steps or to approximate the infinite by the infinite, but in directly constructing, at the definitional level, a structure whose self-consistent extension naturally contains no living space for non-real zeros. That is, the primary prerequisite is the capability to cover the infinite.

Special Emphasis 2

Although the content defined by the Mother Equation carries a certain directionality, its double-reflection symmetry is universal, encompassing both the generality and particularity of various special instances including doubly reflected entire functions, L-functions, and the Riemann Hypothesis. However, from its very inception, no special or direct relation was established with the Riemann zeta function. Indeed, there was a deliberate avoidance of any special or direct relation with the Riemann zeta function, and in fact such a relation is entirely unnecessary. The principle of this paper is to utilize the fact that a counterexample to the Riemann Hypothesis would violate the universality of the Mother Equation, thereby demonstrating that the Riemann Hypothesis is established as a logical fact. That is, the counterexample cannot stand, so the Riemann Hypothesis naturally holds. If one were to force the Mother Equation to establish a direct relation with the Riemann zeta function, this would undoubtedly constitute a typical case of cheating-style customization, violating the most basic principle of reverse derivation and rendering the result of structured reverse derivation objectively invalid. The above confirmation is a prerequisite. Otherwise, one will fall into the set-theoretic inertia of formalized proof, whereas the Mother Equation is actually a definition under structured reverse derivation.

Special Emphasis 3

(1) To ensure the rigor and objectivity of the result of structured reverse derivation, this paper makes no assumptions whatsoever. That is, structured reverse derivation is neither a formalized forward mathematical proof nor a formalized proof by contradiction. Through structured reverse derivation, the self-detonation of contradictory facts is derived. By the invalidity of the counterexample, the Riemann Hypothesis is demonstrated to be instantaneously established as a logical fact.

(2) Defining the Mother Equation as the Whole 1 (not an assumption) has the sole purpose of achieving infinite coverage in directionality. However, one cannot say that because the definition of the Whole 1 happens to align with certain characteristics in directionality, it constitutes a specialized, specifically targeted customization aimed at the Riemann Hypothesis or any other particular object. In summary, this represents a fundamental methodological and conceptual divergence between structured reverse derivation and the traditional formalized proof determination path. That is, the Riemann Hypothesis is not proved. Rather, the counterexample cannot exist within self-consistency due to contradiction, and thus the Riemann Hypothesis is instantaneously established as a logical fact. Therefore, it is fundamentally impossible to use the traditional standard of formalization, which approximates the infinite by the finite, to judge whether a proposition such as the Riemann Hypothesis, which requires infinite coverage, holds.

Special Emphasis 4

(Object Domain Restriction of Unitarity Repulsion and the Interlocking Structure of the Three Gates)

The unitary repulsion mechanism of the Mother Equation is, at the level of its very design, not a universal executioner directed at all doubly reflected entire functions; the configurations it admits are structurally restricted to counterexample configurations carrying nonreal zeros. Every stage of the mechanism, namely the triggering of a configuration's admission, the contradiction self detonation induced by the negative eigenvalue on the χ_{AB} representation, and the final repulsion output (the execution list), acts exclusively on counterexample configurations from beginning to end. It is an execution mechanism, not an admission mechanism.

This gives rise to an object discipline that must be strictly observed: the ξ function is not permitted to enter this repulsion procedure, and has never entered it. The ξ function possesses no counterexample identity; its type of zero configuration lies outside the admitted scope. Hence there exists no interaction whatsoever between the ξ function and unitary repulsion at the judgment level: the relation is neither that of passing nor that of not triggering, but the inapplicability of the procedure. This distinction is decisive: so long as any theoretical interaction at the judgment level between the ξ function and the repulsion procedure remains conceivable (even one whose output is positive energy), the last space for a customization accusation persists. The inapplicability of the procedure cuts this relation off entirely: the legitimacy of the ξ function is inferred exclusively from the completeness of the repulsion outcome, without passing through any judgment action that takes the ξ function as its object. In other words, every judgment of the repulsion procedure is applied only to counterexample configurations; the legitimacy of the ξ function is a logical consequence of the global fact of repulsion completeness, not the result of any individual judgment.

Once repulsion is complete, every member of F that has not been repelled inherits the legitimacy of its zero configuration by default (not being on the execution list constitutes the certificate of passage). This is no special exemption but the logical necessity of repulsion completeness: when all configurations carrying nonreal zeros have been excluded wholesale at the definitional level, the zero distributions of the remaining members unfold naturally as the self consistency extension of the Whole 1, and lie inherently within the legitimate configurations of the stable spectrum. Just as covering the infinite must be achieved in one stroke through a structural constraint on the infinite totality, so the confirmation of legitimacy can only be given in the global form of repulsion completeness, never in

the finite form of judgments given object by object.

A typical conceptual substitution must here be refuted: namely, misreading unitary repulsion as a universal executioner applied to all qualifiers admitted at the first round, and then demanding to know why these qualifiers cannot alike undergo the execution. This demand presupposes that the repulsion mechanism has adjudicating authority over all members of F; yet the admitted objects are restricted at the level of design to counterexample configurations, and counterexample identity is the sole qualification for entering the repulsion procedure. The ξ function does not possess this identity and has never been submitted to the procedure; the egalitarian question frame simply does not arise here. If a critic's point of attack is placed on how the ξ function could pass the repulsion procedure, the object of argument has already been substituted in advance. Special emphasis, once more: this paper never claims that the ξ function passes the repulsion procedure (the second gate); what it claims is that all counterexample configurations self detonate at the execution, and that the ξ function, as an F member that has not been repelled, inherits legitimacy by default. The former is an admission judgment with the ξ function as its object (constituting customization); the latter is a repulsion judgment with the entire class of counterexample configurations as its object (constituting coverage). The difference between the two is structural.

From the above object discipline, the full text forms an interlocking structure of three gates. The first gate is the three entry conditions (entireness, Schwarz reflection, strict central symmetry without multiplier), deliberately loose as sufficient conditions; its function is to cover the infinite: the constraint spectrum must encompass all doubly reflected entire functions, and the width of the entrance is not a defect but the coverage strategy itself. The second gate is the unitarity checkpoint: as an execution mechanism it kills counterexample configurations exclusively, closing the living space of infinitely many nonreal zeros in one stroke through the invariant of representation theory, the eigenvalue $-C$ on the χ_{AB} representation. The third gate is the forward derivation: its function is not a second trial but an audit, answering the question "after the annihilation of all counterexamples, does a surviving path genuinely exist?" The audit checklist consists of existence (strict positive definiteness $\{C+\epsilon, \epsilon\}$ of the binary collapsed orbit at $\sigma=1/2$), uniqueness (the critical line as the unique stable output manifold of the entire complex plane), and the absence of intermediate states (the discrete jumps of stabilizer orders; no transition zone exists between positive and negative definiteness). If one perceives a crack between the execution of the counterexamples and the legitimacy of the ξ function, the forward derivation is precisely the weld point of that crack: it independently confirms that the stable manifold pointed to by the repulsion outcome objectively exists and is unique, so that the reverse conclusion "all counterexamples are annihilated" receives forward support, and the bidirectional closure is thereby formed.

Preliminary Legitimacy Argument: The Historical Identity Between the Mother Equation and Riemann's Original Construction

(A Preliminary Note on Historical Tracing, Physical Legitimacy by Citation, Structural Identity, and Methodological Legitimacy)

Based on an analysis of Riemann's 1859 original German paper **Über die Anzahl der Primzahlen unter einer gegebenen Grösse** (Monatsberichte der Berliner Akademie, 1859, S. 671–680) ^{[1]671, 1} three mutually interlocking settings of the original paper can be confirmed:

(1) Object locking. In the original text, after presenting the series representation and functional equation of $\zeta(s)$, Riemann introduced a prefactor to strictly construct the regularized function. Upon completing the variable substitution $s = \frac{1}{2} + it$, Riemann introduced the core object $\xi(t)$ with the definite article die Function, with the

¹ B. Riemann, Über die Anzahl der Primzahlen unter einer gegebenen Grösse, Monatsberichte der Berliner Akademie, 1859, S. 671–680, cited here from S. 671.

original text reading: "Ich bezeichne nun mit $\xi(t)$ die Function ... welche sich leicht als eine ganze Function von t^2 ausdrücken lässt." ^{[1]672} This object is given by the following explicit definition:

$$\xi(t) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s), \quad s = \frac{1}{2} + it$$

Regarding the distribution of zeros, the original German text states: "und es ist sehr wahrscheinlich, dass alle Wurzeln reell sind." In context, this statement clearly refers to the roots (**Wurzeln**) of the equation $\xi(t) = 0$. As $\zeta(s)$ has a pole at $s = 1$, it was never incorporated by Riemann into this zero-statement; later textbooks reformulated this proposition as the set of "non-trivial zeros" of $\zeta(s)$, but the object of the German original remains strictly anchored at $\xi(t)$.

(2) Holistic declaration. In the original text, after constructing $\xi(t)$, Riemann pointed out that this function is "**für alle endlichen Werthe von t endlich**" (finite for all finite values of t), and can be expanded into a rapidly convergent power series in t^2 ("**lässt sich nach Potenzen von t^2 in eine sehr schnell convergirende Reihe entwickeln**"). That is, the original text states: "**Diese Function ist für alle endlichen Werthe von t endlich, und lässt sich nach Potenzen von t^2 in eine sehr schnell convergirende Reihe entwickeln.**" ^{[1]672} This is equivalent to declaring that $\xi(t)$ is an entire function, analytic on the whole complex plane without singularities. But $\zeta(s)$ has a singularity at $s = 1$ and does not possess this entireness.

(3) Position binding. In the original text, Riemann transformed the zero problem of $\xi(t)$ into an equation in the auxiliary variable t through the substitution $s = \frac{1}{2} + it$. The German term "reell" in the expression "**alle Wurzeln reell**" directly corresponds to t being real, i.e., $\text{Im}(t) = 0$. From the algebraic relation $s = \frac{1}{2} + it$, $t \in \mathbb{R}$ is strictly equivalent to $\text{Re}(s) = 1/2$. Therefore, through this variable substitution and the German expression "**reell**", the original text intrinsically binds the geometric position of the roots to the line $\text{Re}(s) = 1/2$ in the very act of writing. It should be clarified that Riemann's original formulation of this statement is "**sehr wahrscheinlich**" (very likely), and he immediately adds: "**Hiervon wäre allerdings ein strenger Beweis zu wünschen**" ^{[1]672} (a rigorous proof of this would be desirable), indicating that this is a conjecture posed in the original text rather than an established conclusion. Any attempt to study zeros while ignoring or removing the prefactor h(s) or shifting to the ζ -function path does not conform to the object of Riemann's original German text.

Section 1. The German Original Source of the Riemann Hypothesis and the Strict Definition of the ξ Function

Riemann's original paper, submitted to the Berlin Academy in 1859, bears the German title and source:

Über die Anzahl der Primzahlen unter einer gegebenen Grösse²

Monatsberichte der Berliner Akademie, 1859, S. 671–680^{[1]671}. Collected in: Gesammelte Werke, Teubner, Leipzig, 1876, S. 136; 2. Aufl. 1892, S. 145. In this paper, the core object Riemann introduced is not the Riemann zeta function $\zeta(s)$ of today's standard textbooks, but the Riemann xi function, strictly constructed through regularization by a prefactor. Its strict definition in the original text is:

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s) \quad (1)$$

² B. Riemann, Über die Anzahl der Primzahlen unter einer gegebenen Grösse, Monatsberichte der Berliner Akademie, 1859, S. 671–680; Gesammelte Werke, Teubner, Leipzig, 1876, S. 136; 2. Aufl. 1892, S. 145.

that is, in the original text, $(\xi(t) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s), s = \frac{1}{2} + it)$.

From this definition, $\xi(s)$ (that is, $\xi(t)$ in the original) is analytic on the entire complex plane $\mathbb{C}$ and is an entire function (**ganze Funktion**)^{[1]672}; its only singularities are precisely cancelled by the prefactor $s(s-1)$ and the poles of $\Gamma(s/2)$. When $s \in \mathbb{R}$, it follows directly from (1) that $\xi(s) \in \mathbb{R}$, that is, ξ takes real values on the real axis.

Riemann explicitly established the following foundational properties in the original text:

(a) Entireness: $\xi(s)$ is an entire function, analytic on the whole complex plane $\mathbb{C}$ without singularities.

(b) Reality on the real axis: when $s \in \mathbb{R}$, it follows directly from (1) that $\xi(s) \in \mathbb{R}$.

From (a), (b), and the identity theorem for analytic functions, the Schwarz reflection symmetry follows immediately:

$$\xi(\overline{s}) = \overline{\xi(s)} \quad (2) \quad [5]$$

(c) Functional equation: in the original text, through the modular transformation of the Jacobi theta function

$$\vartheta(\tau) = \sum_{n=-\infty}^{\infty} e^{-\pi n^2 \tau}, \quad \vartheta(\tau) = \frac{1}{\sqrt{\tau}} \vartheta\left(\frac{1}{\tau}\right) \quad (3)$$

Riemann strictly derived that $\xi(s)$ satisfies

$$\xi(s) = \xi(1-s) \quad (4) \quad [1]672$$

This identity shows that $\xi(s)$ is centrally symmetric about the point $s = \frac{1}{2}$.

From (2) and (4) one further rigorously deduces: the zero set of $\xi(s)$ is symmetric about both the real axis $\text{Im}(s)=0$ and the critical line $\text{Re}(s) = 1/2$. It was precisely on the basis of these global symmetries that Riemann stated his famous conjecture in the original text: all zeros of $\xi(s)$ lie on the critical line $\text{Re}(s) = 1/2$.

Furthermore, when stating the conjecture in the original text, Riemann directly connected it to the zero distribution of the above entire function, in the German formulation:

"und es ist sehr wahrscheinlich, dass alle Wurzeln reell sind." (and it is very likely that all roots are real.)

[Note: the original text uses the variable t , where $s = \frac{1}{2} + it$; here it is transcribed into the modern standard notation of complex analysis as s , the mathematical substance being completely identical.]

Here "reell" (real) refers to the auxiliary variable t introduced by Riemann through the variable substitution. Restoring the condition that t be real to the variable s yields $\text{Re}(s) = 1/2$. This conclusive evidence of the German original shows that Riemann's original conjecture was, from the very moment of its formulation, strictly bound to the zeros of the entire function $\xi(s)$, and not to the "nontrivial zeros" of $\zeta(s)$, a later equivalent paraphrase. The former is the essential object; the latter is merely a projection at the set theoretic level.

Section 2. The ξ Function and the ζ Function: Equivalence of Zeros, Inequivalence of Structures

In the standard formulation, the Riemann Hypothesis is often paraphrased as "all nontrivial zeros of $\zeta(s)$ lie on the critical line $\text{Re}(s) = 1/2$ "^[7]. This paraphrase is fully equivalent to Riemann's original statement, and the equivalence is strictly guaranteed by the following classical fact:

Lemma (correspondence of the zeros of ξ and ζ). Let $s \in \mathbb{C}$ with $0 < \text{Re}(s) < 1$ (that is, in the critical strip).

Then the prefactor $\frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)$ has no zeros in the critical strip (its zeros lie only at $s = 0, 1$ and at the negative even poles of $\Gamma(s/2)$, none of which are in the critical strip) [2]. Therefore:

$$\xi(s) = 0 \iff \zeta(s) = 0 \quad (\text{within the critical strip } 0 < \text{Re}(s) < 1)$$

Proof sketch: follows directly from (1) and the analytic properties of the prefactor. Since the prefactor has no zeros in the critical strip and is not identically zero, ξ and ζ have exactly the same zero set in the critical strip (counted with multiplicities).

Corollary: Riemann's original conjecture that "all zeros of ξ lie on $\text{Re}(s) = \frac{1}{2}$ " is equivalent to the standard formulation "all nontrivial zeros of $\zeta(s)$ lie on $\text{Re}(s) = \frac{1}{2}$ ". However, the equivalence of the zero sets guarantees only the identity of the conclusions at the set theoretic level, and in no way guarantees the identity of the objects at the level of structural mathematics. The following gives a strict proof of inequivalence within the framework of the Mother Equation.

The space $\mathcal{F}$ of doubly reflected entire functions defined by the Mother Equation has the following entry conditions (sufficient conditions):

- (1) Entireness: f is an entire function, that is, $f \in \mathcal{O}(\mathbb{C})$, analytic on the whole plane without singularities;
- (2) Schwarz reflection symmetry: $f(\bar{s}) = \overline{f(s)}$, $\forall s \in \mathbb{C}$;
- (3) Functional equation (central symmetry): $f(s) = f(1-s)$, $\forall s \in \mathbb{C}$.

Theorem (structural inequivalence of ξ and ζ). $\xi(s) \in \mathcal{F}$, while $\zeta(s) \notin \mathcal{F}$.

The proof verifies this in two strict steps.

Step one: inspection of entireness. By (a) of Section 1 and Riemann's original German text, $\xi(s)$ is an entire function and satisfies condition (1). However, $\zeta(s)$ has a pole of first order at $s=1$:

$\zeta(s) \sim \frac{1}{s-1}$ ($s \rightarrow 1$) [2]; hence $\zeta(s)$ is not an entire function and violates condition (1). This single item already suffices to determine that $\zeta(s) \notin \mathcal{F}$.

Step two: independent inspection of the double reflection symmetry. Even if we provisionally ignore the pole at $s=1$ (for example, by examining the situation within the critical strip $0 < \text{Re}(s) < 1$), $\zeta(s)$ still fails condition (3). It is true that $\zeta(s)$ satisfies a generalized functional equation with the nontrivial multiplier $\chi(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s)$ [2]:

$$\zeta(s) = \chi(s) \zeta(1-s)$$

However, the multiplier $\chi(s) \not\equiv 1$ here is precisely the proof that $\zeta(s)$ by itself possesses no intrinsic, multiplier free, strict central symmetry $f(s) = f(1-s)$. The entry condition of the Mother Equation demands strict symmetry without multiplier, and $\zeta(s)$ is hereby strictly excluded. Indeed, from $\xi(s) = h(s)\zeta(s)$, where $h(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)$ and $\xi(s) = \xi(1-s)$, one obtains $\zeta(s) = \frac{h(1-s)}{h(s)} \zeta(1-s)$. Clearly $\frac{h(1-s)}{h(s)} \not\equiv 1$ (for example, taking $s = \frac{1}{2} + it$, this ratio is a complex phase factor of modulus 1, but is not identically equal to 1). Therefore: $\zeta(s) \neq \zeta(1-s)$ (in general). $\zeta(s)$ fails condition (3).

Furthermore, although from $\xi(\bar{s}) = \overline{\xi(s)}$ and $h(\bar{s}) = \overline{h(s)}$ (the coefficients of the prefactor are all real, π is real, and the Γ function takes real values on the real axis and satisfies Schwarz reflection) one can formally derive that

$\zeta(\bar{s}) = \overline{\zeta(s)}$ holds wherever $h(s) \neq 0$, this "inherited" reflection symmetry is borrowed, packaged, and external. It depends on the existence of the prefactor $h(s)$, and is not an intrinsic attribute of $\zeta(s)$ as an independent mapping. Once detached from the "packaging" of $h(s)$, $\zeta(s)$ by itself generates no geometric reflection whatsoever about the critical line.

Conclusion: $\xi(s)$ satisfies all entry conditions of $\mathcal{F}$ naturally, from the level of its very definition; $\zeta(s)$, by reason of the pole and of $f(s) \neq f(1-s)$, is strictly excluded from $\mathcal{F}$.

The precise mathematical formulation of **"a tiny deviation leads to a thousand mile error"**

The above theorem reveals the exact location of the detonation point: detonation point = the introduction or removal of the prefactor $h(s)$. Riemann's "tiny" difference of introducing $h(s)$ (a multiplication or division by one prefactor) causes, at the level of structural mathematics, a "thousand mile" separation:

For $\xi(s)$: multiplying by $h(s)$ cancels the pole and simultaneously encodes the double reflection symmetry intrinsically, making it a natural geometric object under the action of the V_4 group, falling into the constraint spectrum of the Mother Equation.

For $\zeta(s)$: after removing $h(s)$, the pole is exposed again, and the functional equation degenerates into a relation with a nontrivial multiplier. $\zeta(s)$ falls from a natural element of the framework of the Mother Equation to an external, incomplete object.

Therefore, "studying the Riemann Hypothesis with the ζ function" amounts, within the framework of the Mother Equation, to "using an object that itself lacks double reflection symmetry to derive a proposition about a doubly reflected symmetric structure". This is not an "equivalent substitution", but a structural misalignment of tool and object.

Section 3. From Riemann's Original Functional Equation to Double Reflection Symmetry: The Strict Generation of the V_4 Group

The functional equation (4) and the Schwarz reflection (2) in Riemann's original paper are, mathematically, equivalent to a doubly reflected symmetric system. The strict derivation follows:

Step 1 (Schwarz reflection).

Define the reflection map about the real axis:

$$\iota B(s) = \bar{s} \quad (5)$$

By (2), ξ under the action of ιB satisfies:

$$\xi(\iota B(s)) = \xi(\bar{s}) = \overline{\xi(s)} \quad (6)$$

Step 2 (the functional equation as central symmetry).

Define the central symmetry map about the point $s = \frac{1}{2}$:

$$\iota C(s) = 1 - s \quad (7)$$

By (4), ξ under the action of ιC satisfies:

$$\xi(\iota C(s)) = \xi(1 - s) = \xi(s) \quad (8)$$

Step 3 (generating the reflection about the critical line).

Composing (5) and (7), define:

$$\iota A(s) = (\iota B \circ \iota C)(s) = \overline{1-s} = 1-\bar{s} \quad (9)$$

This ιA is the strict geometric reflection about the line $Re(s) = 1/2$.

Verifying the transformation of ξ under ιA : replacing s by $\bar{s}$ in (4) gives $\xi(\bar{s}) = \xi(1-\bar{s})$; combining with (2), one has:

$$\xi(\iota A(s)) = \xi(1-\bar{s}) = \xi(\bar{s}) = \overline{\xi(s)} \quad (10)$$

Step 4 (closure of the V_4 group structure).

Verify that ιA and ιB are both involutions and commute with each other:

$$\begin{aligned} \iota A(\iota A(s)) &= 1 - \overline{(1-\bar{s})} = s, \quad \iota B(\iota B(s)) = \bar{\bar{s}} = s \\ \iota A(\iota B(s)) &= 1 - \bar{\bar{s}} = 1-s = \iota C(s), \quad \iota B(\iota A(s)) = \overline{1-\bar{s}} = 1-s = \iota C(s) \end{aligned}$$

Hence $\{\text{id}, \iota A, \iota B, \iota C\}$ forms the Klein four group V_4 , and $\iota C = \iota A \circ \iota B = \iota B \circ \iota A$.

This is not a piece of group theoretic decoration attached by human hand, but the strict mathematical growth of formulas (1)–(4) in Riemann's original text.

Section 4. Structural Identity of the Mother Equation and Riemann's Original Construction

Comparing the entry conditions of the Mother Equation item by item with the mathematical structures in Riemann's original text of 1859:

Element of the Mother Equation	Direct counterpart in Riemann's original text (1859)	Relation
Entire function f	$\xi(s)$ is entire (1)(a)	Same object
Channel B: $\iota_B(s) = \bar{s}$	Schwarz reflection symmetry (2)(5)(6)	Same structure
Channel A: $\iota_A(s) = 1 - \bar{s}$	Composition of the functional equation (4) with Schwarz reflection (2), (9)(10)	Same structure
Composite reflection $\iota_C(s) = 1 - s$	Central symmetry expressed by the functional equation (4), (7)(8)	Same structure
V_4 group $\{\text{id}, \iota_A, \iota_B, \iota_C\}$	Klein four group generated by ι_A, ι_B	Same structure
Critical line $Re(s) = 1/2$	Fixed point set of ι_A , that is, the solution of $s = 1 - \bar{s}$	Same geometric object
Real axis $Im(s) = 0$	Fixed point set of ι_B , that is, the solution of $s = \bar{s}$	Same geometric object
Unique intersection point $s = 1/2$	Geometric intersection of the critical line and the real axis	Same geometric object

Conclusion: The Mother Equation does not erect an "external framework" beyond Riemann's original construction. The entry conditions of the Mother Equation (double reflection symmetry) and the original mathematical structures Riemann set up for the ξ function in 1859 (functional equation + Schwarz reflection) are, mathematically, different formulations of the same structure. From the level of its definition onward, Riemann's ξ function naturally satisfies all sufficient conditions for entering the constraint spectrum of the Mother Equation.

The criticism, found in the formalization tradition, that "the Mother Equation needs to establish an external bridge

to the Riemann Hypothesis" rests on the following double misplacement:

Misplacement one: substitution of the object. This criticism mistakes the ζ function of today's standard textbooks for the original object of the Riemann Hypothesis, and thereby demands a proof of an "inclusion relation" between the Mother Equation and the ζ function. However, by the lemma of Section 2, the nontrivial zeros of the ζ function and the zeros of the ξ function are in one to one correspondence within the critical strip; and by Sections 3 and 4, the ξ function naturally falls into the doubly reflected symmetric spectrum of the Mother Equation from the level of Riemann's original definition of 1859. Therefore, the correct path of association is:

$$\text{Mother Equation} \Rightarrow \text{Doubly Reflected Entire Functions} \Leftarrow \xi \text{ function (Riemann's Original Object)} \Leftrightarrow \text{Non-trivial Zeros of } \zeta(s)$$

Misplacement two: reversal of direction. As a universal constraint structure for doubly reflected entire functions, the logical direction of the Mother Equation is "objects satisfying double reflection symmetry naturally fall into the spectrum". The ξ function satisfies $f(s) = f(1-s)$ and $f(\bar{s}) = \overline{f(s)}$; hence it is, in itself, exactly the kind of object the Mother Equation describes. This is not "the Mother Equation contains the ξ function", but "the structural skeleton of the ξ function and the definitional level of the Mother Equation are completely identical".

If one follows the formalistic demand to "prove that the Mother Equation contains the ζ function", one falls into a methodological paradox: once the Mother Equation is degraded into a "set to be proved" and the ζ function degraded into an "element to be verified", one has already fallen into the set theoretic inertia of formalization; whereas the actual position of the Mother Equation is the Whole 1 of a self consistency structure, whose constraint spectrum unfolds without depending on the "joining" of any external object to establish its validity.

Section 5. The Legitimacy Root of Physical Self Detonation: The Holistic Physical Genes in Riemann's Original Construction

$$\xi(t) = \underbrace{\frac{1}{2} s(s-1) \pi^{-s/2} \Gamma\left(\frac{s}{2}\right)}_{h(s)} \cdot \zeta(s)$$

On the basis of the above historical and strictly mathematical facts, the structures adopted in the "physical self detonation" part of this paper, namely the dual channel unitary system, the V_4 group geometry, the reflection axes, and the critical line, are not a "physical analogy" imposed from outside upon a purely number theoretic object. The following gives a strictly one to one mathematical physics correspondence chain:

(1) The functional equation (4) and the Jacobi modular transformation (3).

Physically, the modular transformation of the Jacobi theta function, $\vartheta(\tau) = \frac{1}{\sqrt{\tau}} \vartheta(1/\tau)$, corresponds to duality in statistical mechanics: the equivalence of the high temperature and the low temperature descriptions; in string theory, it corresponds to T duality, the equivalence of the compactification radius R and $1/R$. Its mathematical essence is an involutive map $\iota C(s) = 1-s$, generated in the Mother Equation by the dual channel coupling.

(2) Schwarz reflection (2).

Physically, the Schwarz reflection $\xi(\bar{s}) = \overline{\xi(s)}$ corresponds to time reversal symmetry (T symmetry) or to the real field condition (physical observables must be real). Its mathematical essence is the reflection map $\iota B(s) = \bar{s}$, constituting Channel B of the Mother Equation.

(3) Composition of the functional equation with Schwarz reflection (9).

The composition of the two independent reflection channels generates the reflection $\iota A(s) = 1 - \bar{s}$ about the critical line $\text{Re}(s) = \frac{1}{2}$, constituting Channel A of the Mother Equation. This is not a construction by human hand, but the necessary geometric product of the two original reflections.

(4) The V_4 group structure.

The Klein four group generated by ιA and ιB is not algebraic decoration attached by human hand, but the necessary mathematical growth of double reflection symmetry (see Step 4 of Section 3). Physically, it corresponds to the complete symmetry group of a system, whose irreducible representations $(\chi_0, \chi_A, \chi_B, \chi_{AB})$ directly determine the stable modes and the unstable modes of the system.

(5) The critical line $\text{Re}(s) = \frac{1}{2}$ as the unique stable output.

It is not an external parameter, but the unique geometric intersection of the fixed point set of ιA with the fixed point set of ιB (the real axis). Physically, this corresponds to the critical point of a phase transition: when $\sigma = \frac{1}{2}$, the four element orbit collapses into a two element orbit, the energy of the χ_{AB} mode is $2\varepsilon > 0$, and the structure is stable; when $\sigma \neq \frac{1}{2}$, the four element orbit expands, the energy of the χ_{AB} mode is $-4(C - \varepsilon) < 0$, and the structure self detonates in contradiction at the definitional level. The critical line is precisely the critical point of this phase transition.

(6) The unitarity axiom $\forall \delta\Phi \neq 0: \Delta E > 0$

It corresponds to the strict convexity constraint of the doubly reflected symmetric structure at the level of energy stability. Physically, unitarity means that the response of the system to any nonzero perturbation must be positive, that is, the system is in a strictly stable state. Once a nonreal zero configuration activates the χ_{AB} mode, it produces negative energy $-4(C - \varepsilon) < 0$, forming an irreconcilable contradiction self detonation with unitarity at the definitional level. This corresponds to the prohibition of spontaneous symmetry breaking in physics: the system does not permit the existence of configurations that destroy the consistency of the dual channel coupling.

(7) The delta kernel coupling and the inter channel coupling strength C .

The delta kernels $K_A = C\delta(w - (1 - \bar{s}))$ and $K_B = C\delta(w - \bar{s})$ in the Mother Equation correspond physically to the minimal realization of point to point coupling: reflection symmetry is realized as enforced coupling rather than as a continuous interaction. This is the minimal realization of double reflection symmetry, introducing no additional structural assumptions.

(8) The regularization parameter ε and the intrinsic boundary $C > \varepsilon$.

Physically, ε corresponds to the local fluctuation gap, and C corresponds to the inter channel coupling strength. The condition $C > \varepsilon$ is the definitional boundary of the very concept of a dual channel system: if $\varepsilon \geq C$ the system degenerates into two non communicating single channels and is no longer the object defined by the Mother Equation. This is analogous to the requirement in fluid mechanics that viscosity must lie below a threshold, otherwise the system behaves as a solid rather than as a fluid.

Section 6. The Legitimacy of the Mother Equation, the Source of Physical Self Detonation, and the Structural Ceiling of the Zeta Path

From the strict arguments of the above three dimensions, the following conclusions follow.

(1) The essential object of the Riemann Hypothesis is the ξ function, not the ζ function.

In the original German text of 1859, Riemann explicitly constructed $\xi(s)$ to the standard of a "ganze Funktion" and bound its zero distribution by "alle Wurzeln reell". From the very moment of its birth, this problem was a proposition of global self consistency concerning a doubly reflected symmetric entire function. $\zeta(s)$ is only the incomplete projection of this whole structure after the removal of a specific prefactor.

(2) "A tiny deviation leads to a thousand mile error" has been strictly proved.

The introduction and removal of the prefactor $h(s)$ is a "tiny" matter at the level of the set theory of zeros (a nonzero factor does not change the zeros), but a "thousand mile" matter at the level of structural mathematics: it determines whether the object belongs to the space $\mathcal{F}$ of the Mother Equation, whether the $V4$ group geometry grows intrinsically, and whether the physical self detonation mechanism applies.

(3) The Mother Equation needs no "external bridge" to the Riemann Hypothesis.

The formalization tradition demands "prove that the Mother Equation contains the Riemann Hypothesis"; this criticism rests on a double misplacement. First, it mistakes the ζ function of standard textbooks for the original object of the Riemann Hypothesis, thereby demanding that the Mother Equation accommodate an external element that itself lacks double reflection symmetry. Second, it reverses the logical direction between the Mother Equation and the ξ function: the Mother Equation is a universal constraint structure for doubly reflected entire functions, and the ξ function, from the level of Riemann's definition of 1859, naturally satisfies all sufficient conditions for entering this constraint spectrum.

Therefore, the Mother Equation is not an external framework for the Riemann Hypothesis, but the direct explicitization of the definitional level structure of Riemann's original ξ function. Between the two there exists no bridge that needs to be "proved", only a structural identity that needs to be recognized.

(4) The legitimacy of physical self detonation lies in the return to the source, not in cross domain analogy.

The structures adopted in the "physical self detonation" part of this paper, namely the dual channel unitary system, the $V4$ group geometry, the reflection axes, and the critical line, are not a "physical analogy" imposed from outside upon a purely number theoretic object. Every structural element of the Mother Equation, duality (the functional equation), time reversal (Schwarz reflection), the phase transition critical point (the critical line), and the unitarity stability constraint, corresponds directly to the mathematical symmetries already intrinsically present in formulas (1)–(4) of Riemann's original text.

The success of physical self detonation and of mathematical self detonation does not consist in "applying" physics as an external tool to a number theoretic problem, but in releasing, in the modern language of mathematical physics, the double reflection symmetry, the involutive maps, and the holistic self consistency constraints already encoded but never explicitized in Riemann's original construction of 1859. This is a return to this holistic self consistency, not a crossing of domains.

(5) Even a "success" starting from the ζ function would not genuinely take down the original Riemann Hypothesis.

By the lemma of Section 2, the zeros of ξ and of ζ correspond within the critical strip; hence "all nontrivial zeros of ζ lie on the critical line" is indeed equivalent at the set theoretic level to "all zeros of ξ lie on the critical line". This equivalence guarantees that a proof completed from the ζ function is correct in its conclusion. However, the correctness of the conclusion does not equal the identity of the object.

The original proposition written by Riemann in the German text of 1859 has as its object $\xi(s)$, a complete structure possessing the entirety of a "ganze Funktion" and intrinsic double reflection symmetry. The original content of the

Riemann Hypothesis includes not only the set theoretic fact that "the zeros lie on the critical line", but, further, the structural necessity of "why they must lie on this line": namely, the V_4 group geometry, the unitarity constraint, and the holistic self consistency forcibly grown from the double reflection symmetry.

The ζ function is only an equivalent paraphrase of this original proposition after the removal of a specific prefactor. It preserves the set theoretic information of the zeros, but loses the entirety of the whole function, loses the strict central symmetry $f(s) = f(1 - s)$, and loses the intrinsic skeleton of the V_4 group geometry. Therefore:

the original Riemann Hypothesis (concerning ξ) is a structural mathematical proposition about the holistic self consistency of doubly reflected symmetric entire functions; the zero proposition of the ζ function is the projection of the above original proposition at the set theoretic level.

The two stand in a relation of containment and contained: the original proposition contains the zero distribution, but the zero distribution does not contain the structural core of the original proposition. Proving the projection correct does not equal restoring the original object that was projected. Therefore, suppose a path based on the ζ function eventually "succeeds"; its epistemological position is that of proving a set theoretic proposition equivalent in conclusion to the original proposition, without returning to the holistic self consistency of Riemann's original construction: it does not explain why the critical line is the unique one, does not release the intrinsic double reflection symmetry of the ξ function, and does not achieve structural coverage of the infinite zero set at the definitional level.

In summary, through the prefactor association, the ξ framework naturally contains the zero information of ζ , but the ζ framework does not contain the structural information of ξ (such as entirety, the V_4 group geometry, the unitarity constraint, and so on); that is, the ξ function and the ζ function stand in a relation of one directional containment. According to the German original source, the Riemann Hypothesis should be repositioned and returned from "an asymptotic problem of the zero distribution of the ζ function" to "a structural fact of the holistic self consistency of doubly reflected symmetric entire functions". These are not different solutions of the same problem, but lines of thought for two different problem forms. To take down the Riemann Hypothesis means that one must return to Riemann's original object $\xi(s)$, and close the living space of all nonreal zeros in one stroke through unitarity self detonation within the framework of the Mother Equation. The solution of the Riemann Hypothesis is not the setting of an external goal of conquest, but should be restored to an internal logical fact, restored to that original proposition about doubly reflected symmetric entire functions which Riemann wrote down in 1859.

Introduction

The Riemann Hypothesis asserts that all nontrivial zeros of the Riemann zeta function lie on the critical line $Re(s) = 1/2$. Since its formulation in 1859, all existing research paths on this proposition concerning the global distribution of an infinite zero set, whether the numerical verification of the first tens of trillions of zeros^[8] or the continual improvement of lower bounds for proportions of zeros and of asymptotic estimates in analytic number theory^[6], share at the methodological level one and the same essence: implementing step by step approximation to the infinite whole by finite steps. However, the extrapolation of finite data cannot logically constitute a genuine coverage of the infinite zero set; however far the computational range is expanded, or however much the estimate constants are optimized, the epistemological gap between stepwise approximation and the global assertion remains unbridgeable. Therefore, the premise for solving the Riemann Hypothesis "is not, as in traditional methods, a reliance on ever finer approximation of the infinite by the finite", but a fundamental reversal of methodology: one must first achieve a global structural coverage of the infinite zero set, that is, an infinite coverage.

Accordingly, this paper adopts a strategy of structured reverse derivation. This strategy does not rely on the extrapolation or superposition of local data; instead, by constructing a dual channel system with an intrinsic unitarity

constraint, the Mother Equation, it embeds the global constraint on zero configurations directly into the structural definition. This Mother Equation is not a reverse customization aimed at the specific properties of the Riemann zeta function; it is an algebraic geometric framework growing forward from the abstract definition of double reflection symmetry. On the configuration space of the complex plane, the reflection $\iota A(s) = 1 - \bar{s}$ about the line $Re(s) = 1/2$ and the reflection $\iota B(s) = \bar{s}$ about the real axis are both involutive maps and commute with each other; the group they generate is exactly the Klein four group $V4 = \{id, \iota A, \iota B, \iota C\}$. The kernel structure of the Mother Equation is explicitly defined by the dual channel coupling as:

$$\mathcal{K}(s, w) = \frac{1}{2}(\mathcal{K}_A + \mathcal{K}_B) + \varepsilon \delta(s - w), \mathcal{K}_A = C \delta(w - (1 - \bar{s})), \mathcal{K}_B = C \delta(w - \bar{s}),$$

$$E[\Phi] = \iint_{X \times X} \mathcal{K}(s, w) \Phi(s) \Phi(w) d\lambda(s) d\lambda(w).$$

The unitarity axiom requires that for all $\forall \delta \Phi \neq 0, \Delta E[\delta \Phi] > 0$. This axiom, together with the very concept of "dual channel", yields the intrinsic condition $C > \varepsilon \geq 0$. This is not an adjustable inequality, but the definitional boundary of the very concept of a dual channel system: if the local fluctuation ε dominates the inter channel coupling C , the system degenerates into two non communicating single channel structures and is no longer the object defined by the Mother Equation. From this, the geometric structure of the Mother Equation naturally grows two reflection axes L_A and L_B on the complex plane, and their unique intersection point $s = 1/2$ is not an external parameter, not the result of numerical fitting, but the necessary product of the dual channel definition in geometry.

It must be specially pointed out that the Mother Equation here is not an equation to be solved, nor a set or function space containing particular solutions; it is defined as a Whole 1, an indivisible structural constraint whole. Within this framework, "1" is not a number, but refers to the following: double reflection symmetry as the sufficient condition for entering the constraint spectrum of this Whole; the unitarity axiom as the internal repulsion mechanism of the Whole; the critical line $Re(s) = 1/2$ as the unique stable output after repulsion completeness of the Whole; the three together constitute the indivisible structural constraint whole. As a Whole, the Whole 1 contains the objects of double reflection symmetry and all constraint spectra unfolded by it; hence zero configurations naturally obey the self consistency rules of the Whole 1, that is, they are defined as a self consistency structure (not a hypothesis).

On this basis, this paper establishes a key dimension reduction result of independent mathematical value (Theorem 9.3A). This theorem states: any doubly reflected entire function f satisfying the functional equation $f(s) = f(1 - s)$ and the Schwarz reflection $f(\bar{s}) = \overline{f(s)}$ admits a unique representation $f(s) = g((s - \frac{1}{2})^2)$, where g is an entire function with real coefficients. This theorem strictly reduces the two dimensional distribution problem of infinitely many zeros on the complex plane to the one dimensional zero problem of the real entire function $g(w)$, and gives an unavoidable classification: the negative real zeros of g correspond to zeros on the critical line, the positive real zeros correspond to zeros on the real axis, and the nonreal zeros correspond strictly to configurations deviating from the critical line. Therefore, within this framework, the Riemann Hypothesis is equivalent to the assertion: all zeros of $g(w)$ are negative real numbers. This is not a simplifying trick, but a structural necessity forcibly imposed by double reflection symmetry. Once one enters this soil of symmetry, entire functions must grow in the form $g((s - \frac{1}{2})^2)$; no other form is possible.

The core mechanism of the Mother Equation lies in the global structural constraint imposed by its unitarity axiom. This axiom requires that the energy response of the system to any nonzero local perturbation be strictly positive, that is, the energy functional is strictly convex at the qualitative level. Restricting the Mother Equation to the four element orbit $\mathcal{O}(s_0) = \{s_0, s_1, s_2, s_3\}$ generated by the dual channel coupling, the energy functional degenerates into a quadratic form, and its geometric kernel K_{geom} has a definite spectral decomposition over the irreducible

representations of the group algebra of V_4 . In particular, on the representation χ_{AB} , which is simultaneously antisymmetric under the two reflection channels, the eigenvalue of K_{geom} is $-C$, where $C > 0$ is the channel coupling strength. This negative sign is an invariant of representation theory: whether computed through the 4×4 matrix of the four element orbit, or through the eight dimensional gradient space after the first order Jet lift, or through the infinite dimensional limit of heat kernel regularization, the eigenvalue on the χ_{AB} subspace is invariably $-C$. No regularization operation or dimensional lift can change this sign.

When one considers the zero configuration of the field on the four element orbit and carries out the first order Jet expansion, this negative eigenvalue is inherited by the gradient space, causing the energy contribution of the χ_{AB} mode to be $-(C - \varepsilon)|w|^2$. Under the conceptual survival condition $C > \varepsilon$ of the Mother Equation, the energy of the χ_{AB} mode is strictly negative, directly forming a contradiction self detonation with the unitarity axiom. This self detonation is not the conflict of an external hypothesis, but the direct logical consequence of the definitional level self consistency of the Mother Equation. When $\sigma = \frac{1}{2}$, the four element orbit collapses into a two element orbit, the energy of the χ_{AB} mode is $2\varepsilon > 0$, and the structure is stable; when $\sigma \neq \frac{1}{2}$, the four element orbit expands into a four dimensional one, the energy of the χ_{AB} mode is $-4(C - \varepsilon) < 0$, and the structure self detonates in contradiction at the definitional level. The critical line $\text{Re}(s) = 1/2$ is precisely the critical point of this phase transition: there exist no intermediate states and no approximately stable regions; once a configuration with nonreal zeros is incorporated into the framework of the Mother Equation, it immediately forms an irreconcilable contradiction self detonation with the axiom at the definitional level.

Thus, the Mother Equation achieves global coverage of the infinite zero set through a two step compression. In the first step, Theorem 9.3A reduces the zero configurations on the complex plane to the zero distribution of the real entire function $g(w)$. In the second step, the unitarity axiom of the Mother Equation excludes, in one stroke at the structural level, the configurations with nonreal zeros. Combining this with the strict positivity of the Riemann ξ function on the real interval $(0, 1)$ (a classical fact of number theory), the zeros on the real axis are further compressed to the negative half axis, and thereby correspond strictly to the critical line. Two steps of compression: the structure excludes the nonreal numbers, number theory excludes the positive real numbers; that is, two independent mechanisms logically complement each other and the gap is closed. Under this double repulsion, the existence space of nonreal zeros has been completely excluded by the structure; once their configuration enters the framework of the Mother Equation, it immediately triggers a contradiction self detonation. This mechanism does not rely on point by point enumeration of the zero distribution or on finite extrapolation; its efficacy covers the entire infinite zero set.

The whole argument proceeds from zero assumptions. To cover the infinite completely, the Mother Equation is defined as the Whole 1; as a Whole, this Whole 1 naturally contains double reflection symmetry and all constraint spectra unfolded by it, and is not specially customized to any particular property of the Riemann zeta function; $1/2$ is a geometric intersection point, not a parameter; Theorem 9.3A is a universal dimension reduction theorem, not a customized tool for ζ ; unitarity repulsion applies to all doubly reflected entire functions, not to ζ alone. The non formalized stance of this paper is: rather than attempting to re derive every step within the framework of formalized proof, it establishes closure at the level of logical facts. The Riemann zeta function is introduced only at the end of the argument, as one of the objects possessing double reflection symmetry; its zero distribution is constrained by the structural repulsion of the Mother Equation.

The repulsion of the infinite set of nonreal zeros renders the establishment of the Riemann Hypothesis a logical fact consequence of definitional level self consistency, rather than something achievable by the superposition of finite approximations. This is precisely the epistemological response of structured reverse derivation to the problem of infinite coverage: replacing extrapolation of local data by global structural coverage, and replacing point by point

approximative verification by contradiction self detonation at the definitional level.

The organization of the full text is as follows : The part on physical contradiction self detonation (Parts One to Eight) establishes the construction of the Mother Equation, the dual channel geometry, the four element orbits, the matrix representations, and the complete computation of unitarity self detonation. Part Nine, the mathematical self detonation, gives the complete proof of Theorem 9.3A, the classification of zeros, the χ_{AB} structure of local germs, the well definedness of the Jet lift, and the contradiction chain of Theorem 9.5, and completes the locking of the critical line in combination with the positivity of the ξ function on the real axis. Part Ten, the forward derivation, unfolds the Mother Equation as the Whole 1 in the forward direction: the forward protocol, the structural growth of the reflection axes, the complete scan of the energy landscape, the phase transition theorem, and the unique stable output theorem. Part Eleven proceeds from structural growth to the locking of the zeros: the dimension reduction of Theorem 9.3A, the classification of zeros, the structural exclusion of the nonreal zeros, and the locking of the critical line through real axis positivity. Part Twelve unifies the three parts and forms the bidirectional closure: the forward growth theorem, the bidirectional equivalence theorem, the structural rigidity of the bidirectional closure, the immediate logical fact, and the final conclusion. Finally, the paper discusses the epistemological implications of this methodology for the problem of infinite coverage.

Main Text

Volume One: The Successful Physical Contradiction Self Detonation

Part One: The Construction and Structural Definition of the Mother Equation

1.1 Why Two Channels?

Observation: the mathematical structure of any system obeying two reflection symmetries at once necessarily involves two independent reflection operations.

Define reflection A as the mirror reflection about the line $\text{Re}(s) = \frac{1}{2}$:

$$\iota_A(s) = 1 - \bar{s}$$

Define reflection B as the mirror reflection about the line $\text{Im}(s) = 0$:

$$\iota_B(s) = \bar{s}$$

Verify the involutive property:

$$\iota_A(\iota_A(s)) = 1 - \overline{(1 - \bar{s})} = 1 - (1 - s) = s, \quad \iota_B(\iota_B(s)) = \bar{\bar{s}} = s$$

Both are involutions.

Verify the commutation:

$$\iota_A(\iota_B(s)) = 1 - \bar{\bar{s}} = 1 - s, \quad \iota_B(\iota_A(s)) = \overline{1 - \bar{s}} = 1 - s$$

Hence the two commute, $\iota_A \circ \iota_B = \iota_B \circ \iota_A = \iota_C$, where the composite reflection is $\iota_C(s) = 1 - s$. The generated group $\{\text{id}, \iota_A, \iota_B, \iota_C\}$ forms the Klein four group V_4 .

The construction motive of the Mother Equation: to realize the two generating reflections of V_4 as the two coupled channels of a system.

1.2 The Strict Definition of the Mother Equation (Not a Hypothesis)

The configuration space is defined as $X = \mathbb{C}$ (the complex plane).

The field $\Phi: X \rightarrow \mathbb{R}$ is a real valued field; this requirement stems from the real axis symmetry of ι_B .

The two kernel structure is defined as

$$\mathcal{K}(s, w) = \frac{1}{2}(\mathcal{K}A(s, w) + \mathcal{K}B(s, w)) + \varepsilon \cdot \delta(s - w),$$

where the channel A kernel $\mathcal{K}A$ enforces the coupling of s with $\iota_A(s) = 1 - \bar{s}$:

$$\mathcal{K}A(s, w) = \begin{cases} C & \text{if } w = 1 - \bar{s} \\ 0 & \text{otherwise} \end{cases}$$

and the channel B $\mathcal{K}B$ enforces the coupling of s with $\iota_B(s) = \bar{s}$:

$$\mathcal{K}B(s, w) = \begin{cases} C & \text{if } w = \bar{s} \\ 0 & \text{otherwise} \end{cases}$$

The coupling strength $C > 0$ is a classical geometric coupling parameter. The regularization parameter $\varepsilon \geq 0$ is the local fluctuation gap. $\delta(s - w)$ guarantees that the self coupling of the field at its own point is nonnegative.

The global energy functional is defined as

$$E[\Phi] = \iint X \times X \mathcal{K}(s, w) \Phi(s) \Phi(w) d\lambda(s) d\lambda(w),$$

where λ is a finite reference measure on X , ensuring that the integral is well defined.

The unitarity axiom (the intrinsic axiom of the Mother Equation) is defined as

$$\eta \equiv 1 \Leftrightarrow \forall \delta\Phi \neq 0: \Delta E[\delta\Phi] > 0,$$

that is, the energy response of the system to any nonzero perturbation is strictly positive, and the system is strictly convex at the level of energy.

1.3 The Structural Definition of the Mother Equation

The Mother Equation is not an equation to be solved, nor a set or function space containing particular solutions. It is defined (not hypothesized) as a Whole 1, a self consistency structure with global constraint power.

Within this framework, "1" is not a number, but an indivisible structural constraint whole:

- (1) double reflection symmetry is the sufficient condition for entering the constraint spectrum of this Whole;
- (2) the unitarity axiom is the internal repulsion mechanism (execution mechanism) of the Whole;
- (3) the critical line $Re(s) = 1/2$ is the unique stable output after repulsion completeness of the Whole.

The division into "sufficient condition" and "repulsion mechanism (execution mechanism)" above corresponds directly to the subsequent complete logical chain of this paper, namely the dimension reduction of Theorem 9.3A and Theorem 9.5, which together constitute the full coverage from the entry conditions to the repulsion output. The former locks the structural form of the zeros, the latter cuts off the living space of the nonreal zeros.

Therefore, when the Riemann ξ function is placed later under the examination of the Whole 1, what is being stated is not the set theoretic proposition that " ξ belongs to a certain subset", but the structural proposition that " ξ satisfies double reflection symmetry, and hence its zero distribution, as the natural unfolding of the structure of the Whole 1, is endogenous to that self consistency rule". As the Whole 1, the Mother Equation does not depend on the "joining" of any external object to establish its validity; as a Whole, the Whole 1 naturally contains the objects of double

reflection symmetry and their constraint spectra, and the structural possibilities of zero configurations are themselves the natural unfolding of the definition of the Whole 1, one of the many structural forms it contains, and not a special pointing.

Definition (the Mother Equation as the Whole). The Mother Equation is defined as a Whole 1, a self consistency structure with global constraint power. Double reflection symmetry, the unitarity axiom, and the critical line together constitute the indivisible structural constraint whole.

Although the starting point of the argument of this paper is the definition of the Mother Equation itself, and not any particular property of the Riemann zeta function. ζ is introduced only later as a concrete instance satisfying double reflection symmetry. This paper is a structured reverse derivation, not a proof, and the standards and requirements of the two differ in essence. In summary, not only was the construction of the Mother Equation earlier than any subsequent step concerning ζ , there is also no suspicion of circularity of "customizing the Mother Equation in order to prove the Riemann Hypothesis".

Part Two: The Geometric Growth of the Reflection Axes and the Intersection Point

2.1 The Reflection Axes Growing from the Kernel Definition

The fixed point set of channel A satisfies $s = \iota_A(s) = 1 - \bar{s}$. Writing $s = \sigma + i\gamma$, one has $\sigma + i\gamma = 1 - (\sigma - i\gamma) = 1 - \sigma + i\gamma$, which solves to $\sigma = \frac{1}{2}$.

Hence the fixed point set of channel A is the critical line $L_A = \{s: \text{Re}(s) = \frac{1}{2}\}$.

The fixed point set of channel B satisfies $s = \iota_B(s) = \bar{s}$, that is, $\sigma + i\gamma = \sigma - i\gamma$, which solves to $\gamma = 0$.

Hence the fixed point set of channel B is the real axis $L_B = \{s: \text{Im}(s) = 0\}$.

2.2 The Intersection of the Two Axes (a Pure Geometric Fact)

Solving for the intersection $L_A \cap L_B$, that is, $\text{Re}(s) = 1/2$ together with $\text{Im}(s) = 0$, gives the unique intersection point $s = \frac{1}{2}$.

This number $1/2$ is not an external parameter, not the result of numerical fitting. It is the unique point grown by the dual channel definition of the Mother Equation in the geometry of the complex plane.

2.3 The Key Geometric Observation

If $s \in L_A$ (that is, $\text{Re}(s) = 1/2$), then

$$\iota_A(s) = 1 - \bar{s} = 1 - (\frac{1}{2} - i\gamma) = \frac{1}{2} + i\gamma = s,$$

so s is a fixed point under channel A.

At the same time, $\iota_B(s) = \bar{s} = \frac{1}{2} - i\gamma \neq s$ ($\gamma \neq 0$), so s is not a fixed point under channel B, but its image $\bar{s}$ still lies on L_A , since $\text{Re}(\bar{s}) = \frac{1}{2}$. Therefore, the critical line L_A is the fixed point set of channel A, and also the invariant set of channel B: ι_B maps L_A to itself.

Part Three: Perturbation Analysis of the Four Element Orbit Expansion

3.1 Definition of the Perturbation

Consider a local perturbation $\delta\Phi$ supported at the point $s_0 = \sigma + i\gamma$, where $\gamma > 0$.

From the definition of the kernel, the forced coupling mechanism of the Mother Equation follows directly:

Channel	Action of the kernel	Forced partner point
A	$\mathcal{K}A(s_0, w) \neq 0 \Leftrightarrow w = 1 - \bar{s}0$	$s_1 = 1 - \sigma + i\gamma$
B	$\mathcal{K}B(s_0, w) \neq 0 \Leftrightarrow w = \bar{s}0$	$s_2 = \sigma - i\gamma$

3.2 Closure of the Four Element Orbit

The partner points s_1 and s_2 each continue to generate partners under the two channels.

From $s_1 = 1 - \sigma + i\gamma$:

channel A acts to give $\iota_A(s_1) = 1 - \bar{s}1 = 1 - (1 - \sigma - i\gamma) = \sigma + i\gamma = s_0$ (returning to the origin);

channel B acts to give $\iota_B(s_1) = \bar{s}1 = 1 - \sigma - i\gamma = s_3$ (a new point).

From $s_2 = \sigma - i\gamma$:

channel A acts to give $\iota_A(s_2) = 1 - \bar{s}2 = 1 - (\sigma + i\gamma) = 1 - \sigma - i\gamma = s_3$ (a new point);

channel B acts to give $\iota_B(s_2) = \bar{s}2 = \sigma + i\gamma = s_0$ (returning to the origin).

From $s_3 = 1 - \sigma - i\gamma$:

channel A acts to give $\iota_A(s_3) = 1 - \bar{s}3 = 1 - (1 - \sigma + i\gamma) = \sigma - i\gamma = s_2$;

channel B acts to give $\iota_B(s_3) = \bar{s}3 = 1 - \sigma + i\gamma = s_1$.

The orbit closes. The action of the two channels on s_0 generates the closed four element set

$$\mathcal{O}(s_0) = \{s_0, s_1, s_2, s_3\} = \{\sigma + i\gamma, 1 - \sigma + i\gamma, \sigma - i\gamma, 1 - \sigma - i\gamma\}.$$

3.3 The Group Theoretic Root of the Orbit Dimension (the Orbit Stabilizer Theorem)

The above two orbit structures, the four element one and the two element one, are not numerical coincidences, but can be derived strictly from the group action of V_4 . V_4 acts on an arbitrary point $s \in \mathbb{C}$; the orbit is $\mathcal{O}(s) = V_4 \cdot s$, and the stabilizer is $\text{Stab}(s) = \{g \in V_4 : g \cdot s = s\}$. By the orbit stabilizer theorem,

$$|\mathcal{O}(s)| = \frac{|V_4|}{|\text{Stab}(s)|} = \frac{4}{|\text{Stab}(s)|}$$

If $s \notin L_A \cup L_B$, then $\iota_A(s) \neq s$ and $\iota_B(s) \neq s$, hence $\text{Stab}(s) = \{\text{id}\}$ and $|\mathcal{O}(s)| = 4$. This is the four element orbit.

If $s \in L_A$ but $s \notin L_B$ (that is, $\sigma = \frac{1}{2}$, $\gamma \neq 0$), then $\iota_A(s) = s$ while $\iota_B(s) \neq s$, hence $\text{Stab}(s) = \{\text{id}, \iota_A\}$ and $|\mathcal{O}(s)| = 2$. This is the two element collapsed orbit.

If $s \in L_A \cap L_B$ (that is, $s = \frac{1}{2}$), then $\text{Stab}(s) = V_4$ and $|\mathcal{O}(s)| = 1$. This is the fixed point.

Therefore, the distinction between the four element orbit and the two element orbit is an intrinsic structure of the group action, and does not depend on point by point verification.

3.4 The Two Geometric Cases

Case one: $\sigma = \frac{1}{2}$ (on the intersection line L_A).

In this case,

$$s_0 = \frac{1}{2} + i\gamma,$$

$$s_1 = 1 - \frac{1}{2} + i\gamma = \frac{1}{2} + i\gamma = s_0 \leftarrow \text{collapse!}$$

$$s_2 = \frac{1}{2} - i\gamma,$$

$$s_3 = 1 - \frac{1}{2} - i\gamma = \frac{1}{2} - i\gamma = s_2 \leftarrow \text{collapse!}$$

The four element orbit collapses into the two element orbit $\mathcal{O}(s_0) = \{s_0, s_2\}$, where $s_0 \neq s_2$ (since $\gamma > 0$). The geometric reason is that $s_0 \in L_A$, hence $\iota_A(s_0) = s_0$, and the fixed point property of channel A collapses s_1 into s_0 .

Case two: $\sigma \neq \frac{1}{2}$ (deviating from the intersection line).

In this case, $s_1 = 1 - \sigma + i\gamma \neq s_0$ ($1 - \sigma \neq \frac{1}{2}$), $s_2 = \sigma - i\gamma \neq s_0$ (since $\gamma \neq 0$), and $s_1 \neq s_2$ (different real parts). The four element orbit remains four dimensional, and the four points are pairwise distinct.

Part Four: The Matrix Representation (Reproducible Element by Element)

4.1 The Discretization Restriction and the Decoupling Between Orbits

Restricting the Mother Equation to the four element orbit $\mathcal{O}(s_0)$, define the field values $\Phi_i = \Phi(s_i)$, $i = 0, 1, 2, 3$. The energy functional degenerates into the quadratic form

$$E = \sum_{i,j} K_{ij} \Phi_i \Phi_j,$$

where $K_{ij} = \mathcal{K}(s_i, s_j)$.

Strict account of the decoupling between orbits. By the definition of the kernel, $\mathcal{K}(s, w) \neq 0$ only when $w \in \{s, \iota_A(s), \iota_B(s)\}$. For any s_0 , its orbit $\mathcal{O}(s_0)$ is closed under ι_A and ι_B ; if $w \notin \mathcal{O}(s_0)$, then $\mathcal{K}(s_i, w) = 0$ holds for all $s_i \in \mathcal{O}(s_0)$. Hence there is no coupling between different orbits, and the stability analysis can be carried out independently on a single orbit. This establishes that a 4×4 matrix suffices to determine the local stability of the continuous system in the neighborhood of this orbit.

4.2 The Element by Element Computation of the Channel A Kernel $\mathcal{KA}$

$\mathcal{KA}(s_i, s_j) = C$ if and only if $s_j = \iota_A(s_i) = 1 - \bar{s}_i$.

Verify all 16 matrix elements one by one			
(i,j)	compute $1 - \bar{s}_i$	equals s_j ?	result
(0,0)	$1 - (\sigma - i\gamma) = 1 - \sigma + i\gamma$	$\neq \sigma + i\gamma$ (since $\sigma \neq \frac{1}{2}$)	0
(0,1)	$1 - \sigma + i\gamma$	$= s_1$	C
(0,2)	$1 - \sigma + i\gamma$	$\neq \sigma - i\gamma$	0
(0,3)	$1 - \sigma + i\gamma$	$\neq 1 - \sigma - i\gamma$ (since $\gamma \neq 0$)	0
(1,0)	$1 - (1 - \sigma - i\gamma)$	$= \sigma + i\gamma = s_0$	C
(1,1)	$\sigma + i\gamma$	$\neq 1 - \sigma + i\gamma$	0

(1,2)	$\sigma + i\gamma$	$\neq \sigma - i\gamma$	0
(1,3)	$\sigma + i\gamma$	$\neq 1 - \sigma - i\gamma$	0
(2,0)	$1 - (\sigma + i\gamma) = 1 - \sigma - i\gamma$	$\neq \sigma + i\gamma$	0
(2,1)	$1 - \sigma - i\gamma$	$\neq 1 - \sigma + i\gamma$	0
(2,2)	$1 - \sigma - i\gamma$	$\neq \sigma - i\gamma$	0
(2,3)	$1 - \sigma - i\gamma$	$= s_3$	C
(3,0)	$1 - (1 - \sigma + i\gamma) = \sigma - i\gamma$	$\neq \sigma + i\gamma$	0
(3,1)	$\sigma - i\gamma$	$\neq 1 - \sigma + i\gamma$	0
(3,2)	$\sigma - i\gamma$	$= s_2$	C
(3,3)	$\sigma - i\gamma$	$\neq 1 - \sigma - i\gamma$	0

Therefore

$$\mathcal{K}A = C \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

4.3 The Element by Element Computation of the Channel B Kernel $\mathcal{K}B$

$\mathcal{K}B(s_i, s_j) = C$ if and only if $s_j = \iota_B(s_i) = \bar{s}i$.

Verify all 16 matrix elements one by one			
(i, j)	compute $\bar{s}i$	equals s_j ?	result
(0,0)	$\sigma - i\gamma$	$\neq \sigma + i\gamma$	0
(0,1)	$1 - \sigma + i\gamma$	$\neq 1 - \sigma + i\gamma$	0
(0,2)	$\sigma - i\gamma$	$= s_2$	C
(0,3)	$\sigma - i\gamma$	$\neq 1 - \sigma - i\gamma$	0
(1,0)	$1 - \sigma - i\gamma$	$\neq \sigma + i\gamma$	0
(1,1)	$1 - \sigma - i\gamma$	$\neq 1 - \sigma + i\gamma$	0
(1,2)	$1 - \sigma - i\gamma$	$\neq \sigma - i\gamma$	0
(1,3)	$1 - \sigma - i\gamma$	$= s_3$	C
(2,0)	$\sigma + i\gamma$	$= s_0$	C
(2,1)	$\sigma + i\gamma$	$\neq 1 - \sigma + i\gamma$	0
(2,2)	$\sigma + i\gamma$	$\neq \sigma - i\gamma$	0
(2,3)	$\sigma + i\gamma$	$\neq 1 - \sigma - i\gamma$	0
(3,0)	$1 - \sigma + i\gamma$	$\neq \sigma + i\gamma$	0
(3,1)	$1 - \sigma + i\gamma$	$= s_1$	C

(3,2)	$1 - \sigma + i\gamma$	$\neq \sigma - i\gamma$	0
(3,3)	$1 - \sigma + i\gamma$	$\neq 1 - \sigma - i\gamma$	0

Therefore

$$\mathcal{K}B = C \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

4.4 The Total Kernel (Purely Geometric Part)

$$K_{\text{geom}} = \frac{1}{2}(\mathcal{K}A + \mathcal{K}B) = \frac{C}{2} \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}$$

Part Five: The Necessity of Regularization and the Intrinsic Scale

5.1 The Singularity of the Purely Geometric Kernel

When $\sigma = \frac{1}{2}$, one has $s_0 = s_1$ and $s_2 = s_3$. In this case, the restrictions of $\mathcal{K}A$ and $\mathcal{K}B$ to the collapsed subspace are

$$\mathcal{K}A|_{\text{sub}} = C \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathcal{K}B|_{\text{sub}} = C \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad K_{\text{geom}}|_{\text{sub}} = \frac{C}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Its eigenvalues are $\lambda_1 = C$ (the symmetric mode (1,1)) and $\lambda_2 = 0$ (the antisymmetric mode (1,-1)). A zero eigenvalue exists, that is, $\ker K_{\text{geom}} \neq \{0\}$.

5.2 The Requirement of the Unitarity Axiom

The axiom of the Mother Equation requires $\eta \equiv 1$, that is, strict positive definiteness. The zero eigenvalue violates this axiom; therefore a regularization term εI must be introduced to eliminate the zero mode:

$$K = K_{\text{geom}} + \varepsilon I = \frac{C}{2} \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix} + \varepsilon \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

5.3 The Scale Relation Between ε and C (the Intrinsic Boundary Condition)

Definition (the intrinsic boundary of a dual channel system). The Mother Equation is defined as a dual channel unitary system. Dual channel means that there exists a nontrivial coupling between the two reflection channels. The relative scale of the parameters ε and C is not an external hypothesis, not a parameter fitted from experimental data, but an intrinsic boundary condition given by the definition of the Mother Equation itself.

The argument is as follows. If $\varepsilon \geq C$, then the local self coupling ε dominates the inter channel coupling C, the system degenerates into two nearly independent single channel systems, the correlation between channel A and channel B is submerged by the local fluctuation, the dual channel structure loses its meaning, and the Mother Equation degenerates into two non communicating mirrors.

Therefore, the self definition of the Mother Equation as a dual channel unitary system intrinsically requires that the inter channel coupling must dominate the local fluctuation, that is,

$$C > \varepsilon \geq 0$$

This is analogous to the situation in defining a fluid: the viscosity must lie below a certain threshold, otherwise the system behaves as a solid. This is not an additional hypothesis of the fluid, but the definitional boundary of the very concept of a fluid.

Part Six: Group Representations and Mode Selection

6.1 The Action of the V_4 Group

On the four element orbit $\mathcal{O}(s_0) = \{s_0, s_1, s_2, s_3\}$, the group $V_4 = \{\text{id}, \iota_A, \iota_B, \iota_C\}$ acts as follows:

Group element	Action	Permutation on the indices $\{0, 1, 2, 3\}$
id	$s \mapsto s$	(0)(1)(2)(3)
ι_A	$s \mapsto 1 - \bar{s}$	(01)(23)
ι_B	$s \mapsto \bar{s}$	(02)(13)
ι_C	$s \mapsto 1 - s$	(03)(12)

6.2 The Irreducible Representations of V_4

V_4 is an abelian group and has four one dimensional irreducible representations:

Representation	$\chi(\text{id})$	$\chi(\iota_A)$	$\chi(\iota_B)$	$\chi(\iota_C)$	Basis vector
χ_0 (trivial)	1	1	1	1	$v_0 = (1, 1, 1, 1)$
χ_A	1	-1	1	-1	$v_A = (1, -1, 1, -1)$
χ_B	1	1	-1	-1	$v_B = (1, 1, -1, -1)$
χ_{AB}	1	-1	-1	1	$v_{AB} = (1, -1, -1, 1)$

6.3 Why Must It Be χ_{AB} ?

Question: through which representation must the perturbation of the Mother Equation go ?

Analyse the physical meaning of each representation:

Trivial representation χ_0 ($v_0 = (1, 1, 1, 1)$) : all four points are in phase. It is invariant under channel A (eigenvalue +1), that is, s_0 and s_1 are in phase; it is invariant under channel B (eigenvalue +1), that is, s_0 and s_2 are in phase. The physical picture is the uniform breathing mode, a global fluctuation of the system.

χ_A ($v_A = (1, -1, 1, -1)$) : antisymmetric under channel A (s_0 and s_1 in antiphase), symmetric under channel B. The physical picture is the destruction of the coupling of channel A while preserving channel B.

χ_B ($v_B = (1, 1, -1, -1)$) : symmetric under channel A, antisymmetric under channel B (s_0 and s_2 in antiphase). The physical picture is the destruction of the coupling of channel B while preserving channel A.

χ_{AB} ($v_{AB} = (1, -1, -1, 1)$) : antisymmetric under channel A (s_0 and s_1 in antiphase), antisymmetric under channel B (s_0 and s_2 in antiphase). The physical picture is the simultaneous destruction of the couplings of both channels.

6.4 The Selection Rule: Why Is the Trivial Representation Forbidden?

If the perturbation goes through the trivial representation $v_0 = (1, 1, 1, 1)$, then the four points are in phase, and both channels A and B are satisfied (in phase coupling has the lowest energy). But this is a uniform translation of the background field, not a local perturbation. What the unitarity axiom requires is the energy of the change relative to the background, and a uniform translation does not change the energy difference.

The deeper reason: the kernel K of the Mother Equation already contains the background coupling. A perturbation must embody the inconsistency relative to the background. The trivial representation is isomorphic to the background and produces no new physical information.

The genuine instability mode must destroy the consistency of both channels at once in order to test the self consistency of the system. Only χ_{AB} challenges both channels simultaneously.

6.5 The Complete Spectral Decomposition of the Total Kernel $\mathcal{K}$ and the Global Stability Picture

Since K_{geom} commutes with all elements of V_4 (verifiable element by element via $P_g K_{\text{geom}} P_g^T = K_{\text{geom}}$, where P_g is the permutation matrix of the group element g), K can be diagonalized simultaneously by the four irreducible representations. Compute all eigenvalues directly:

Representation	Eigenvector	K_{geom}	Eigenvalue $K = K_{\text{geom}} + \varepsilon I$	Sign ($C > \varepsilon$)
χ_0	$v_0 = (1,1,1,1)C$	C	$C + \varepsilon$	> 0
χ_A	$v_A = (1,-1,1,-1)$	0	ε	≥ 0
χ_B	$v_B = (1,1,-1,-1)$	0	ε	≥ 0
χ_{AB}	$v_{AB} = (1,-1,-1,1)$	$-C$	$-C + \varepsilon$	< 0

Global spectral picture. The unitarity axiom requires all eigenvalues to be strictly positive. However, on the four dimensional orbit with $\sigma \neq \frac{1}{2}$, the χ_{AB} mode invariably contributes the negative eigenvalue $-C + \varepsilon < 0$, forming an intrinsic contradiction. This contradiction does not depend on single point hand computation, but is a structural defect of the total kernel K in the complete representation space. This establishes that any perturbation deviating from the critical line triggers a negative energy under the χ_{AB} channel, and is self consistently rejected by the unitarity axiom.

Part Seven: The Energy Computation (Hand Calculated Element by Element)

7.1 Computing Kv_{AB}

Take the vector $v_{AB} = (1, -1, -1, 1)^T$. The total kernel is

$$K = \frac{C}{2} \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix} + \varepsilon \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Action of the geometric part:

$$\begin{aligned} (K_{\text{geom}}v_{AB})_0 &= \frac{C}{2}(0 \cdot 1 + 1 \cdot (-1) + 1 \cdot (-1) + 0 \cdot 1) = -C, \\ (K_{\text{geom}}v_{AB})_1 &= \frac{C}{2}(1 \cdot 1 + 0 \cdot (-1) + 0 \cdot (-1) + 1 \cdot 1) = C, \\ (K_{\text{geom}}v_{AB})_2 &= \frac{C}{2}(1 \cdot 1 + 0 \cdot (-1) + 0 \cdot (-1) + 1 \cdot 1) = C, \\ (K_{\text{geom}}v_{AB})_3 &= \frac{C}{2}(0 \cdot 1 + 1 \cdot (-1) + 1 \cdot (-1) + 0 \cdot 1) = -C. \end{aligned}$$

Hence $K_{\text{geom}}v_{AB} = (-C, C, C, -C)^T$.

Action of the regularization part: $\varepsilon I \cdot v_{AB} = (\varepsilon, -\varepsilon, -\varepsilon, \varepsilon)^T$.

Total action:

$$Kv_{AB} = (-C + \varepsilon, C - \varepsilon, C - \varepsilon, -C + \varepsilon)^T$$

7.2 Computing the Quadratic Form $\langle vAB, KvAB \rangle$

$$\begin{aligned}
 \langle vAB, KvAB \rangle &= v_0(Kv)0 + v_1(Kv)1 + v_2(Kv)2 + v_3(Kv)3 \\
 &= 1 \cdot (-C + \varepsilon) + (-1) \cdot (C - \varepsilon) + (-1) \cdot (C - \varepsilon) + 1 \cdot (-C + \varepsilon) \\
 &= (-C + \varepsilon) + (-C + \varepsilon) + (-C + \varepsilon) + (-C + \varepsilon) \\
 &= -4C + 4\varepsilon
 \end{aligned}$$

7.3 Determination of the Sign

By the intrinsic boundary condition $C > \varepsilon$ of Part Five:

$$\langle vAB, KvAB \rangle = -4C + 4\varepsilon = -4(C - \varepsilon) < 0,$$

strictly negative.

Part Eight: Triggering of the Physical Contradiction Self Detonation

8.1 Derivation of the Contradiction

Mathematical setting. The following energy variation examines second order stability about the zero background field $\Phi_{bg} \equiv 0$, that is,

$$\Delta E[\delta\Phi] = E[0 + \delta\Phi] - E[0] = \langle \delta\Phi, K\delta\Phi \rangle,$$

and the first order term vanishes because the background field is zero.

The axiom of the Mother Equation (not to be violated):

$$\eta \equiv 1 \Leftrightarrow \forall \delta\Phi \neq 0: \Delta E = \langle \delta\Phi, K\delta\Phi \rangle > 0.$$

The actual result:

$$\langle vAB, KvAB \rangle = -4(C - \varepsilon) < 0.$$

Contradiction: there exists a nonzero perturbation vAB whose energy is strictly negative, violating the unitarity axiom.

8.2 The Self Consistent Response of the System

Faced with this contradiction, the Mother Equation has two logical possibilities: first, to modify the axiom and allow negative energy, but then the definition of the Mother Equation as a unitary system would be destroyed; second, to reject the perturbation, which then has no legitimate place of existence in the solution space of the Mother Equation.

The choice of the Mother Equation is: the axiom is part of the definition and cannot be modified. Therefore the system must reject the perturbation source that produces negative energy.

8.3 Comparison of Cases and the Phase Transition

Strict treatment of the degenerate case. When $\sigma = \frac{1}{2}$, one has $s_0 = s_1$ and $s_2 = s_3$, and the four element orbit collapses into a two element one. The discretization then automatically degenerates to the number of independent physical points, avoiding double counting. Physically, coincident points contribute no independent degrees of freedom, and the matrix representation is restricted to the subspace of non coincident points.

Case	Geometric position	Orbit dimension	Antisymmetric mode energy	Unitarity
$\sigma = 1/2$	on the intersection line L_A	2 (collapsed)	$2\varepsilon > 0$	satisfied
$\sigma \neq 1/2$	deviating from the intersection line	4 (expanded)	$-4(C - \varepsilon) < 0$	violated

Conclusion of the phase transition. $\sigma = \frac{1}{2}$ is the stable point of the Mother Equation system. Perturbations deviating from this point are self consistently rejected by the unitarity axiom of the Mother Equation.

Volume Two: Mathematical Contradiction Self Detonation

Part Nine: The Successful Mathematical Contradiction Self Detonation

In what follows, the constraint mechanism of the Whole 1 is applied to the elements of $\mathcal{F}$. Here the ξ function is not "embedded" into the Mother Equation; rather, the Mother Equation, as the structural constraint rule of the Whole 1, is imposed on all entire functions satisfying double reflection symmetry, of which the ξ function is only one.

9.1 The V_4 Group Structure Growing from Double Reflection Symmetry

Definition 9.1.1 (the space of doubly reflected entire functions).

Define $\mathcal{F}$ as the set of entire functions $f: \mathbb{C} \rightarrow \mathbb{C}$ satisfying the following two conditions:

(A) Functional equation: $f(s) = f(1-s)$ for all $s \in \mathbb{C}$;

(B) Schwarz reflection: $f(\bar{s}) = \overline{f(s)}$ for all $s \in \mathbb{C}$.

Theorem 9.1.2 (the natural growth of the V_4 group).

Conditions (A) and (B) generate the Klein four group $V_4 = \{\text{id}, \iota_A, \iota_B, \iota_C\}$ on the complex plane, where:

$$\iota_A(s) = 1 - \bar{s} \text{ (reflection about the line } Re(s) = 1/2),$$

$$\iota_B(s) = \bar{s} \text{ (reflection about the real axis),}$$

$$\iota_C(s) = 1 - s = \iota_A \circ \iota_B \text{ (the composite reflection).}$$

Strict verification (hand computed element by element):

Verify that ι_A is an involution:

$$\iota_A(\iota_A(s)) = 1 - \overline{(1 - \bar{s})} = 1 - (1 - s) = s$$

Verify that ι_B is an involution:

$$\iota_B(\iota_B(s)) = \overline{\bar{s}} = s$$

Verify that the two commute:

$$\iota_A(\iota_B(s)) = 1 - \overline{\bar{s}} = 1 - s,$$

$$\iota_B(\iota_A(s)) = \overline{1 - \bar{s}} = 1 - s,$$

$$\iota_A \circ \iota_B = \iota_B \circ \iota_A = \iota_C.$$

The generated group $\{\text{id}, \iota_A, \iota_B, \iota_C\}$ forms an abelian group, isomorphic to V_4 .

9.2 K_{geom} in the Group Algebra of V_4 and Its Eigenvalues

Definition 9.2.1 (the group algebra element).

In the group algebra $R[V4]$ define

$$K_{\text{geom}} = \frac{C}{2}(\iota_A + \iota_B),$$

where $C > 0$ is the channel coupling strength of the Mother Equation.

Theorem 9.2.2 (the spectral decomposition of K_{geom})

$V4$ is an abelian group and has four one dimensional irreducible representations:

Representation	$\chi(\text{id})$	$\chi(\iota_A)$	$\chi(\iota_B)$	$\chi(\iota_C)$
χ_0 (trivial)	1	1	1	1
χ_A	1	-1	1	-1
χ_B	1	1	-1	-1
χ_{AB}	1	-1	-1	1

Derivation: $V4$ is abelian, so all irreducible representations are one dimensional. On a representation χ , the group element ι_A acts as multiplication by $\chi(\iota_A)$ and ι_B acts as multiplication by $\chi(\iota_B)$. Therefore:

$$K_{\text{geom}}|_{\chi} = \frac{C}{2}[\chi(\iota_A) + \chi(\iota_B)]$$

Substituting one by one:

$$\chi_0: \frac{C}{2}(1 + 1) = C$$

$$\chi_A: \frac{C}{2}(-1 + 1) = 0$$

$$\chi_B: \frac{C}{2}(1 - 1) = 0$$

$$\chi_{AB}: \frac{C}{2}(-1 - 1) = -C$$

Conclusion: the eigenvalue of K_{geom} on the χ_{AB} representation is $-C < 0$ (strictly negative).

Matrix verification (already computed element by element in Parts One to Eight):

The matrix representation of $K_{\{\text{geom}\}}$ on the four element orbit $\{s_0, s_1, s_2, s_3\}$ is

$$K_{\text{geom}} = \frac{C}{2} \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}$$

whose characteristic polynomial is $\det(K_{\text{geom}} - \lambda I) = \lambda^2(\lambda - C)(\lambda + C) = 0$, with eigenvalues $\lambda \in \{C, 0, 0, -C\}$, consistent with the group algebra computation.

9.3 The Complete Classification of Doubly Reflected Entire Functions (Theorem 9.3A)

Theorem 9.3A (the core breakthrough).

For any $f \in \mathcal{F}$ (doubly reflected entire function), f admits a unique representation

$$f(s) = g\left(\left(s - \frac{1}{2}\right)^2\right),$$

where $g(w) = \sum_{k=0}^{\infty} a_k w^k$ is an entire function with real coefficients ($a_k \in \mathbb{R}$).

Derivation (step by step, strict):

Step 1: Taylor expansion at $s = \frac{1}{2}$:

$$f(s) = \sum_{n=0}^{\infty} c_n \left(s - \frac{1}{2}\right)^n, \quad c_n = \frac{f^{(n)}\left(\frac{1}{2}\right)}{n!}$$

Step 2: By condition (A), $f(s) = f(1-s)$. Letting $w = s - \frac{1}{2}$, one has $1-s = \frac{1}{2} - w$, hence

$$f\left(\frac{1}{2} + w\right) = f\left(\frac{1}{2} - w\right),$$

that is,

$$\sum_{n=0}^{\infty} c_n w^n = \sum_{n=0}^{\infty} c_n (-w)^n = \sum_{n=0}^{\infty} c_n (-1)^n w^n$$

Comparing coefficients of equal powers:

$$c_n = (-1)^n c_n$$

When n is odd, $c_n = -c_n \Rightarrow c_n = 0$.

Therefore all odd order coefficients vanish, and f is even about $\frac{1}{2}$:

$$f(s) = \sum_{k=0}^{\infty} c_{2k} \left(s - \frac{1}{2}\right)^{2k}$$

Step 3: Let $z = \left(s - \frac{1}{2}\right)^2$ and define

$$g(z) = \sum_{k=0}^{\infty} c_{2k} z^k$$

Then $f(s) = g\left(\left(s - \frac{1}{2}\right)^2\right)$. Since f is entire, g is also entire (analyticity of the composition: $z \mapsto \frac{1}{2} + \sqrt{z}$ is analytic on $\mathbb{C} \setminus (-\infty, 0]$, and the evenness of f guarantees that g is continuous across the branch cut, so that g admits an analytic continuation to an entire function).

Step 4: Deriving the coefficients $a_k = c_{2k} \in \mathbb{R}$.

By condition (B), $f(\bar{s}) = \overline{f(s)}$. Taking $s = \frac{1}{2} + t$ ($t \in \mathbb{R}$), one has

$$f\left(\frac{1}{2} + t\right) = \overline{f\left(\frac{1}{2} + t\right)}$$

so f takes real values on the real axis.

In particular, $f\left(\frac{1}{2}\right) = g(0) = a_0 \in \mathbb{R}$.

For $g(w) = \sum_{k=0}^{\infty} a_k w^k$, taking $w \geq 0$ (a real number), one has

$$g(w) = f\left(\frac{1}{2} + \sqrt{w}\right) \in \mathbb{R}$$

The function g is entire and takes real values on the interval $[0, +\infty)$. By the identity theorem for analytic functions, if a power series takes real values on a real interval, all its coefficients must be real^[5]. Hence $a_k \in \mathbb{R}$.

Step 5: Uniqueness.

If $f(s) = g_1\left(\left(s - \frac{1}{2}\right)^2\right) = g_2\left(\left(s - \frac{1}{2}\right)^2\right)$, then, setting $z = \left(s - \frac{1}{2}\right)^2$, one has $g_1(z) = g_2(z)$ for infinitely many z (in fact for all $z \in \mathbb{C}$). By the identity theorem, $g_1 = g_2$ ^[5].

Corollary 9.3B (classification of the zeros)

For any zero w_0 of $g(w)$, let $z_0 = \sqrt{w_0}$ (principal branch). Then the zeros of $f(s)$ are $s = \frac{1}{2} \pm z_0$.

Case (i): $w_0 \in (-\infty, 0)$ (negative real).

In this case $z_0 = \pm i\gamma$ (purely imaginary, $\gamma = \sqrt{|w_0|} > 0$), hence

$$s = \frac{1}{2} \pm i\gamma$$

and the zero lies on the critical line $\operatorname{Re}(s) = 1/2$.

Case (ii): $w_0 \in (0, +\infty)$ (positive real). In this case $z_0 = \pm\gamma$ (real, $\gamma = \sqrt{w_0} > 0$), hence

$$s = \frac{1}{2} \pm \gamma$$

and the zero lies on the real axis.

Case (iii): $w_0 \notin \mathbb{R}$ (nonreal).

In this case $z_0 = \delta + i\gamma$ ($\delta \neq 0, \gamma \neq 0$), hence

$$s = \frac{1}{2} \pm (\delta + i\gamma)$$

and the zero deviates from the critical line ($\operatorname{Re}(s) = \frac{1}{2} \pm \delta \neq \frac{1}{2}$).

Key equivalence:

the Riemann Hypothesis $\Leftrightarrow$ all zeros of $g(w)$ are negative real numbers ($w \leq 0$)

9.4 The χ_{AB} Structure of the Local Germ Space

For any $f \in \mathcal{F}$, if $\rho = \sigma + i\gamma$ is a zero of f , then by double reflection symmetry the whole quadruple consists of zeros:

$$s_0 = \rho = \sigma + i\gamma$$

$$s_1 = 1 - \bar{\rho} = 1 - \sigma + i\gamma$$

$$s_2 = \bar{\rho} = \sigma - i\gamma$$

$$s_3 = 1 - \rho = 1 - \sigma - i\gamma$$

Theorem 9.4

Define $v = \text{Im}(f)$. Then:

(a) v is of χ_{AB} type:

$$v(\iota_A(s)) = -v(s), \quad v(\iota_B(s)) = -v(s)$$

(b) v vanishes identically on $L_A \cup L_B$:

$$v|_{L_A} \equiv 0, \quad v|_{L_B} \equiv 0$$

(c) if ρ is a simple zero and $\sigma \neq \frac{1}{2}$, then $\nabla v(\rho) \neq 0$.

Derivation:

(a) From conditions (A) and (B):

$$f(\iota_A(s)) = f(1 - \bar{s}) = f(\bar{s}) = \overline{f(s)}$$

(the first step uses (A): $f(1 - \bar{s}) = f(\bar{s})$; the second step uses (B): $f(\bar{s}) = \overline{f(s)}$).

Taking the imaginary part:

$$v(\iota_A(s)) = \text{Im}(\overline{f(s)}) = -\text{Im}(f(s)) = -v(s)$$

Similarly:

$$f(\iota_B(s)) = f(\bar{s}) = \overline{f(s)}$$

$$v(\iota_B(s)) = -v(s)$$

(b) For $s \in L_B$ (the real axis), $s = \bar{s}$, hence

$$v(s) = v(\bar{s}) = -v(s) \Rightarrow 2v(s) = 0 \Rightarrow v(s) = 0$$

For $s \in L_A$ (the critical line), $s = 1 - \bar{s}$, hence

$$v(s) = v(1 - \bar{s}) = v(\iota_A(s)) = -v(s) \Rightarrow 2v(s) = 0 \Rightarrow v(s) = 0$$

(c) If ρ is a simple zero, then $f'(\rho) \neq 0$. By the Cauchy Riemann equations:

$$f'(\rho) = u_x(\rho) + iv_x(\rho) = v_y(\rho) - iu_y(\rho)$$

Hence

$$|f'(\rho)|^2 = u_x^2(\rho) + v_x^2(\rho) = v_y^2(\rho) + u_y^2(\rho) > 0$$

Therefore $\nabla v(\rho) = (v_x(\rho), v_y(\rho)) \neq (0, 0)$

The V_4 Representation on the Eight Dimensional Gradient Space

Define the eight dimensional vector (four orbit points $\times$ 2 components):

$$w = (\partial_x v(s_0), \partial_y v(s_0), \partial_x v(s_1), \partial_y v(s_1), \partial_x v(s_2), \partial_y v(s_2), \partial_x v(s_3), \partial_y v(s_3))$$

Strict derivation of the concrete form of w :

Let $a = \partial_x v(s_0)$ and $b = \partial_y v(s_0)$.

From $v(1-x, y) = -v(x, y)$, differentiating in x :

$$-\partial_x v(1-x, y) = -\partial_x v(x, y) \Rightarrow \partial_x v(1-x, y) = \partial_x v(x, y)$$

Differentiating in y :

$$\partial_y v(1-x, y) = -\partial_y v(x, y)$$

Hence at $s_1 = (1-\sigma, \gamma) = \iota_A(s_0)$:

$$\partial_x v(s_1) = \partial_x v(s_0) = a$$

$$\partial_y v(s_1) = -\partial_y v(s_0) = -b$$

From $v(x, -y) = -v(x, y)$ (action of ι_B), differentiating in x :

$$\partial_x v(x, -y) = -\partial_x v(x, y)$$

Differentiating in y :

$$-\partial_y v(x, -y) = -\partial_y v(x, y) \Rightarrow \partial_y v(x, -y) = \partial_y v(x, y)$$

Hence at $s_2 = (\sigma, -\gamma) = \iota_B(s_0)$:

$$\partial_x v(s_2) = -\partial_x v(s_0) = -a$$

$$\partial_y v(s_2) = \partial_y v(s_0) = b$$

At $s_3 = (1-\sigma, -\gamma) = \iota_C(s_0) = \iota_A(\iota_B(s_0))$:

$$\partial_x v(s_3) = \partial_x v(\iota_A(s_2)) = \partial_x v(s_2) = -a$$

$$\partial_y v(s_3) = -\partial_y v(\iota_A(s_2)) = -\partial_y v(s_2) = -b$$

In summary:

$$w = (a, b, a, -b, -a, b, -a, -b)$$

Strict verification that w is of χ_{AB} type. Action of ι_A : it exchanges (s_0, s_1) and (s_2, s_3) , with Jacobian $\text{diag}(-1, 1)$. In index notation:

position 0 (the x of s_0) $\rightarrow$ position 2 (the x of s_1), sign $+1$ (since $\partial_x v(s_1) = a = +\partial_x v(s_0)$);

position 1 (the y of s_0) $\rightarrow$ position 3 (the y of s_1), sign -1 (since $\partial_y v(s_1) = -b = -\partial_y v(s_0)$).

That is, the action of ι_A on w is:

$$\iota_A(w) = (a, -b, a, b, -a, -b, -a, b) = -w$$

Action of ι_B : it exchanges (s_0, s_2) and (s_1, s_3) , with Jacobian $\text{diag}(1, -1)$.

position 0 (the x of s_0) $\rightarrow$ position 4 (the x of s_2), sign -1 (since $\partial_x v(s_2) = -a$);

position 1 (the y of s_0) $\rightarrow$ position 5 (the y of s_2), sign $+1$ (since $\partial_y v(s_2) = +b$).

That is, the action of ι_B on w is:

$$\iota B(w) = (-a, b, -a, -b, a, b, a, -b) = -w$$

For $\iota C = \iota A \circ \iota B$:

$$\iota_C(w) = \iota_A(\iota_B(w)) = \iota_A(-a, b, -a, -b, a, b, a, -b) = (a, b, a, -b, -a, b, -a, -b) = +w$$

Hence w is a vector of χ_{AB} type.

9.5 The Joint Self Detonation of the Mother Equation and 9.3A (Theorem 9.5)

Lemma 9.5A (the Jet lift of the delta kernel).

The kernel of the Mother Equation is defined as

$$\mathcal{K}(s, w) = \frac{C}{2} [\delta(w - \iota_A(s)) + \delta(w - \iota_B(s))] + \varepsilon \delta(s - w)$$

For any field Φ vanishing on the four element orbit $\mathcal{O}(\rho) = \{s_0, s_1, s_2, s_3\}$ (that is, $\Phi(s_i) = 0$), the first order Jet expansion of the energy functional of the Mother Equation is

$$\Delta E^{(1)}[\Phi] = \sum_{i,j} \tilde{K}_{ij} w_i \cdot w_j$$

where $w_i = \nabla \Phi(s_i)$ is the gradient vector of the field at the point s_i , and $\tilde{K}$ is the energy matrix on the eight dimensional Jet space, whose form coincides with the matrix of the group algebra element $K_{\text{geom}} = \frac{C}{2} (\iota_A + \iota_B)$ under the Jacobian representation.

Expansion (Lemma 9.5A).

Since $\Phi(s_i) = 0$, the field values themselves contribute nothing to the energy. Consider the first order Taylor expansion of Φ near s_i :

$$\Phi(s) = \nabla \Phi(s_i) \cdot (s - s_i) + O(\|s - s_i\|^2) = w_i \cdot (s - s_i) + O(\|s - s_i\|^2)$$

Substituting into the energy functional:

$$E[\Phi] = \iint \mathcal{K}(s, w) \Phi(s) \Phi(w) d\lambda(s) d\lambda(w)$$

Since $\mathcal{K}$ is a delta type kernel, the integral is nonzero only at the coupling points. Consider the contribution of channel A:

$$\mathcal{K}A(s, w) = C \cdot \delta(w - \iota_A(s))$$

At $s = s_i$, one has $w = \iota_A(s_i) = s_i'$ (some orbit point). At this moment:

$$\Phi(s_i) = 0, \quad \Phi(s_i') = 0$$

but the first order expansion gives:

$$\Phi(s) \approx w_i \cdot (s - s_i) \text{ for } s \approx s_i$$

$$\Phi(w) \approx w_i' \cdot (w - s_i') \text{ for } w \approx s_i'$$

From $w = \iota_A(s)$, near $s = s_i$ one has $w - s_i' = \iota_A(s) - \iota_A(s_i) \approx J_A(s - s_i)$, where J_A is the Jacobian matrix of ι_A .

Since $\iota_A(x, y) = (1 - x, y)$, one has

$$J_A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

Therefore:

$$\Phi(w) \approx wi' \cdot J_A(s - s_i) = (J_A^T wi') \cdot (s - s_i)$$

The first order energy contribution of channel A:

$$E_A^{(1)} = C \sum_i \int s \approx s_i wi' \cdot (s - s_i) \cdot (J_A^T wi') \cdot (s - s_i) d\lambda(s)$$

Owing to the locality of the delta kernel, the integral is confined to an infinitesimal neighborhood of s_i . The matrix elements of this quadratic form are jointly determined by J_A and the index permutation.

Similarly, the contribution of channel B involves $J_B = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

Arranging the eight dimensional gradient vector $W = (w_0, w_1, w_2, w_3)$, the energy matrix $\widetilde{K_{\text{geom}}}$ on the Jet space is constructed as follows:

channel A couples (s_0, s_1) and (s_2, s_3) , with Jacobian $J_A = \text{diag}(-1, 1)$;

channel B couples (s_0, s_2) and (s_1, s_3) , with $J_B = \text{diag}(1, -1)$.

In the eight dimensional space, the matrix elements of $\widetilde{K_{\text{geom}}}$ are constituted by the tensor product of the 2×2 Jacobian blocks and the 4×4 permutation structure:

$$\widetilde{K_{\text{geom}}} = \frac{C}{2} [(PA \otimes JA) + (PB \otimes JB)]$$

where PA, PB are the 4×4 permutation matrices (defined in Parts One to Eight), and JA, JB are the 2×2 Jacobian matrices.

Key observation: as a V_4 module, this eight dimensional representation contains the χ_{AB} component in its irreducible decomposition, and the eigenvalue of K_{geom} on the χ_{AB} subspace is still -C (by group algebra theory, the eigenvalue is an invariant of representation theory, independent of the concrete realization).

Concrete verification: take $W = (a, b, a, -b, -a, b, -a, -b)$ (that is, the w of 9.4 extended to the four element gradient), and compute $\widetilde{K_{\text{geom}}} W$:

contribution of channel A ($PA \otimes JA$):

coupling of $(s_0, s_1): JA w_1 = (-a, -b)$, paired with $w_0 = (a, b)$;

coupling of $(s_2, s_3): JA w_3 = (a, -b)$, paired with $w_2 = (-a, b)$;

contribution of channel B ($PB \otimes JB$):

coupling of $(s_0, s_2): JB w_2 = (-a, -b)$, paired with $w_0 = (a, b)$;

coupling of $(s_1, s_3): JB w_3 = (-a, b)$, paired with $w_1 = (a, -b)$.

Combined computation (taking the x component of s_0 as an example):

$$(\widetilde{K_{\text{geom}}}W)_0 = \frac{C}{2}[(-a) + (-a)] = -Ca$$

Similarly:

$$(\widetilde{K_{\text{geom}}}W)_1 = \frac{C}{2}[(-b) + (-b)] = -Cb$$

That is, $\widetilde{K_{\text{geom}}}W = -CW$. (Lemma 9.5A)

In the above lemma, the Jet lift of the delta kernel is a standard operation in physics, but a note on regularization must be added for mathematical strictness. Strictly speaking, the delta kernel should be understood as the ideal limit of the heat kernel regularization (the Gaussian blob):

$$\mathcal{K}_{\mathcal{A}}^{(\alpha)}(s, w) = \frac{C}{\pi\alpha} \exp\left(-\frac{\|w - \iota_A(s)\|^2}{\alpha}\right)$$

At the zero field values $\Phi(si) = 0$, substituting the first order Taylor expansion into the regularized energy functional $E^{(\omega)}[\Phi]$, the gradient quadratic form in the limit $\alpha \rightarrow 0$ has as its dominant term exactly the $\widetilde{K_{\text{geom}}}$ of Lemma 9.5A. The well definedness of this limit is guaranteed by the following fact: when the four element orbit is expanded ($\sigma \neq \frac{1}{2}$, $\gamma > 0$), the distances between orbit points, of order $|1 - 2\sigma|$ and 2γ , are all positive; hence, as long as $\sqrt{\alpha} \ll \min(|1 - 2\sigma|, 2\gamma)$, the Gaussian blobs at the various points do not overlap, and the cross terms vanish in the limit. Thus "directly substituting the first order expansion into the delta kernel" acquires a rigorous functional analytic foundation under the regularization limit, and the group algebra eigenvalue -C remains unchanged in this limit.

Lemma 9.5C (the heat kernel regularization limit).

The following lemma provides the rigorous regularization limit foundation for the Jet lift of the δ kernel in Lemma 9.5A. Lemma 9.5A has established at the formal level the structure of $\widetilde{K_{\text{geom}}}$ and its spectral eigenvalue -C; the present lemma verifies, via heat kernel regularization, that this formal computation is the exact principal term of the heat kernel limit, and not a formal symbolic manipulation.

Let $\mathcal{O} = \{s_0, s_1, s_2, s_3\}$ be the expanded four element orbit ($\sigma \neq \frac{1}{2}$, $\gamma > 0$).

Take $r_0 = \frac{1}{4} \min(\|1 - 2\sigma\|, 2\gamma) > 0$, ensuring that the closed balls $\overline{B}(si, r_0)$ are pairwise disjoint.

Define the smooth field space

$$\mathcal{D} = \{\Phi \in C^\infty(R^2) : \text{supp}(\Phi) \subset \cup_{i=0}^3 \overline{B}(si, r_0), \Phi(si) = 0 \ (i = 0, 1, 2, 3)\}$$

For $\alpha \in (0, r_0^2)$, define the heat kernel regularized geometric energy:

$$E_{\text{geom}}^{(\omega)}[\Phi] = \frac{C}{2} \sum_{i=0}^3 \int_{B(si, r_0)} [\delta^{(\omega)}(\iota_A(s) - siA) \Phi(s) \Phi(\iota_A(s)) + \delta^{(\omega)}(\iota_B(s) - siB) \Phi(s) \Phi(\iota_B(s))] d\lambda(s)$$

where $siA = \iota_A(si)$, $siB = \iota_B(si)$, and $\delta^{(\alpha)}(z) = \frac{1}{\pi\alpha} e^{-\|z\|^2/\alpha}$.

Assertions:

(i) (pointwise limit) For any $\Phi \in \mathcal{D}$, let $W = (\nabla\Phi(s_0), \dots, \nabla\Phi(s_3)) \in \mathbb{R}^8$ (the gradient coordinates have classical meaning for smooth functions). Then

$$\left| \frac{1}{\alpha} E_{\text{geom}}^{(\alpha)}[\Phi] - \langle W, \widetilde{K_{\text{geom}}} W \rangle \right| \leq C_{\text{rem}} \sqrt{\alpha} \|\Phi\| \mathcal{C}^{2^2}$$

where $C_{\text{rem}} > 0$ depends only on C and the orbit geometry $\mathcal{O}$.

(ii) (eigenvalue inheritance) For any $\Phi \in \mathcal{D}$, if its gradient vector W lies on the χ_{AB} irreducible subspace, then

$$\lim_{\alpha \rightarrow 0^+} \frac{1}{\alpha} E_{\text{geom}}^{(\alpha)}[\Phi] = -C|W|^2 < 0$$

Derivation:

Step 1 (Taylor expansion and remainder control). For $\Phi \in \mathcal{D}$, from $\Phi(si) = 0$ and $\Phi \in Cc^\infty$, within $B(si, r_0)$ one has

$$\Phi(s) = \nabla \Phi(si) \cdot (s - si) + R_i(s), \|R_i(s)\| \leq \frac{1}{2} \|\Phi\| \mathcal{C}^2 \|s - si\|^2$$

Write $wi = \nabla \Phi(si)$ and $u = s - si$.

Step 2 (affine exactness of the Jacobians). The reflections $\iota A(x, y) = (1 - x, y)$ and $\iota B(x, y) = (x, -y)$ are both affine maps; hence

$$\iota A(s) = \iota A(si) + JAu, \quad \iota B(s) = \iota B(si) + JBu$$

hold exactly, where $JA = \text{diag}(-1, 1)$ and $JB = \text{diag}(1, -1)$ are both orthogonal matrices.

Step 3 (the principal term of the channel A integral). Consider the coupling from i to iA :

$$IA, i^{(\alpha)} = \frac{C}{2} \int B(si, r_0) \delta^{(\alpha)}(JAu) \Phi(si + u) \Phi(siA + JAu) d\lambda(u)$$

Substituting the Taylor expansion, the principal term is

$$IA, i^{\text{main}} = \frac{C}{2} \int \mathbb{R}^2 \frac{1}{\pi \alpha} e^{-\|u\|^2/\alpha} (wi \cdot u) (wiA \cdot JAu) d\lambda(u)$$

(the integral may be extended to the whole space, since for $\alpha \ll r_0^2$ the Gaussian decay is far faster than the boundary distance).

Using orthogonality $\|JAu\| = \|u\|$ and the Gaussian second moment formula

$$\int \mathbb{R}^2 \frac{1}{\pi \alpha} e^{-\|u\|^2/\alpha} u_p u_q d\lambda(u) = \frac{\alpha}{2} \delta_{pq}$$

one obtains

$$IA, i^{\text{main}} = \frac{C\alpha}{4} wi \cdot (JA^T wiA)$$

Step 4 (remainder estimate). The cross term and remainder contributions satisfy

$$|IA, i^{(\alpha)} - IA, i^{\text{main}}| \leq \frac{C}{2} \|\Phi\| \mathcal{C}^{2^2} \int \mathbb{R}^2 \frac{1}{\pi \alpha} e^{-\|u\|^2/\alpha} \left(\|u\|^3 + \frac{1}{4} \|u\|^4 \right) d\lambda(u) = O(\alpha^{3/2})$$

Concretely, $\int \|u\|^3 \delta^{(\alpha)} d\lambda \sim \alpha^{3/2}$ and $\int \|u\|^4 \delta^{(\alpha)} d\lambda \sim \alpha^2$.

Step 5 (channel B and summation). The same holds for channel B, whose principal term contributes $\frac{C\alpha}{4} w_i \cdot (J_B^T w_i)_B$.

Summing over $i = 0, 1, 2, 3$ and using the construction of $\widetilde{K_{\text{geom}}}$ in Lemma 9.5A:

$$\frac{1}{\alpha} E_{\text{geom}}^{(\alpha)}[\Phi] = \frac{C}{2} \sum_i [w_i \cdot (J_A^T w_i)_A + w_i \cdot (J_B^T w_i)_B] + O(\sqrt{\alpha}) = \langle W, \widetilde{K_{\text{geom}}} W \rangle + O(\sqrt{\alpha})$$

This derives assertion (i).

Step 6 (regularization limit and eigenvalue inheritance). From Step 5, for any fixed $\Phi \in \mathcal{D}$:

$$\lim_{\alpha \rightarrow 0^+} \frac{1}{\alpha} E_{\text{geom}}^{(\alpha)}[\Phi] = \langle W, \widetilde{K_{\text{geom}}} W \rangle$$

where $W = (\nabla \Phi(s_0), \dots, \nabla \Phi(s_3))$. This limit holds pointwise on $\mathcal{D}$, and the group algebra eigenvalue $-C$ is the limit of the spectra of the regularized operators, not merely a formal computation.

If W is a vector of χ_{AB} type, then by Lemma 9.5A the eigenvalue of $\widetilde{K_{\text{geom}}}$ on the χ_{AB} subspace is $-C$; hence:

$$\lim_{\alpha \rightarrow 0^+} \frac{1}{\alpha} E_{\text{geom}}^{(\alpha)}[\Phi] = -C \|W\|^2 < 0$$

This verifies assertion (ii).

Remark on the relation to Lemma 9.5A: Lemma 9.5A defined at the formal level the structure of $\widetilde{K_{\text{geom}}}$ and verified that its eigenvalue on the χ_{AB} subspace equals $-C$. The present lemma further verifies: when the smooth Jet of Φ falls on the χ_{AB} subspace, the α^{-1} limit of the heat kernel regularized energy is exactly the negative quadratic form corresponding to that eigenvalue. Therefore the $-C$ of Lemma 9.5A is not a formal symbol, but the computable principal term of the heat kernel limit. The two lemmas together constitute the rigorous passage from the discrete delta kernel to the continuous energy variation. (Lemma 9.5C)

Theorem 9.5.

For any $f \in \mathcal{F}$ satisfying the unitarity of the Mother Equation (that is, whose perturbation energy satisfies the unitarity axiom $\Delta E > 0$), $g(w)$ cannot have nonreal zeros.

Derivation:

Step 1: Suppose g has a nonreal zero $w_0 = u + iv$ ($v \neq 0$).

Let $z_0 = \sqrt{w_0} = \delta + i\gamma$ (principal branch, $\delta \neq 0$, $\gamma \neq 0$).

Step 2: $f(s) = g(z^2)$ vanishes at $s_k = \frac{1}{2} + z_k$, where:

$$z_0 = \delta + i\gamma$$

$$z_1 = -\bar{z}_0 = -\delta + i\gamma \text{ (action of } \iota_A),$$

$$z_2 = \bar{z}_0 = \delta - i\gamma \text{ (action of } \iota_B),$$

$$z_3 = -z_0 = -\delta - i\gamma \text{ (action of } \iota_C).$$

The four points are pairwise distinct (since $\delta \neq 0$, $\gamma \neq 0$) and form the four element orbit of V_4 .

Step 3: By Theorem 9.4(c), $v = \text{Im}(f)$ has nonzero gradients at the four orbit points.

Construct the eight dimensional gradient vector w (definition in 9.4). From the zero configuration of Step 2, $w \neq 0$.

Step 4: By the strict verification of 9.4, w is a vector of χ_{AB} type.

Step 5: By Lemma 9.5A, the restriction of the energy functional of the Mother Equation to the first order Jet space is the quadratic form:

$$\Delta E^{(1)}[w] = \langle w, \widetilde{K}w \rangle = \langle w, (\widetilde{K}_{\text{geom}} + \varepsilon I)w \rangle$$

Lemma 9.5C has strictly verified: for the expanded four element orbit, the heat kernel regularized geometric energy $E_{\text{geom}}^{(\alpha)}$ satisfies

$$\lim_{\alpha \rightarrow 0^+} \frac{1}{\alpha} E_{\text{geom}}^{(\alpha)}[\Phi] = \langle W, \widetilde{K}_{\text{geom}} W \rangle$$

and this limit is determined pointwise on the smooth field space $\mathcal{D}$, with remainder estimate $O(\sqrt{\alpha})$. Therefore the Jet lift from the delta kernel to the eight dimensional gradient quadratic form in Lemma 9.5A acquires a rigorous functional analytic foundation under the regularization limit; the eigenvalue $-C$ is the limit of the spectra of the regularized operators, not merely a formal computation.

On the χ_{AB} subspace:

$$\widetilde{K}_{\text{geom}}|_{\chi_{AB}} = -C \quad (\text{derived in Lemma 9.5A}),$$

$$\varepsilon I|_{\chi_{AB}} = +\varepsilon$$

Hence:

$$\Delta E^{(1)}[w] = (-C + \varepsilon)\|w\|^2$$

By the intrinsic condition $C > \varepsilon$ of Parts One to Eight:

$$\Delta E^{(1)}[w] = -(C - \varepsilon)\|w\|^2 < 0$$

Step 6: But the unitarity axiom requires:

$$\forall \delta\Phi \neq 0: \Delta E[\delta\Phi] > 0$$

Contradiction: the nonreal zero w_0 induces the four element orbit, activates the Jet vector w of the χ_{AB} mode, and produces a strictly negative energy, violating the unitarity axiom.

Step 7: This case is excluded instantaneously at the structural level. All zeros of $g(w)$ must be real.

Steps 1 to 7 of Theorem 9.5 above have been strictly derived: for any $f \in \mathcal{F}$ satisfying the unitarity of the Mother Equation, the zeros of $g(w)$ can only be real. In other words, the dual channel unitary structure of the Mother Equation has compressed the zeros from the complex plane onto the real axis. To further assert that these real zeros can only be negative real numbers ($w \leq 0$, corresponding to the critical line $\text{Re}(s) = \frac{1}{2}$), additional arithmetic input is required. Step 8 below is the extra number theoretic fact arising when the above general conclusion is applied to the concrete instance of the Riemann ξ function, and is not a conclusion growing from the Mother Equation alone.

Step 8: Combining the known fact (the ξ function is strictly positive on the real interval $(0, 1)$ [2], hence $g(w) > 0$ for $w > 0$), the real zeros of g can only satisfy $w \leq 0$ (negative real numbers).

By Corollary 9.3B(i):

$$w \leq 0 \Leftrightarrow s = \frac{1}{2} \pm i\gamma \text{ (the critical line)}$$

Note 9.5B (generalization to multiple zeros).

For any m fold zero ρ ($m \geq 1$), the conclusion of Theorem 9.5 still holds.

Derivation:

The unitarity axiom of the Mother Equation requires $\Delta E > 0$ for all nonzero perturbations $\delta\Phi/\equiv 0$. Therefore, it suffices to construct any single nonzero perturbation with $\Delta E < 0$ to violate the axiom.

Step 1 (the V_4 decomposition of the m -th order Jet space).

Consider the direct sum of the m -th order symmetric tensors over the four element orbit:

$$J_O^m = \bigoplus_{i=0}^3 \text{Sym}^m(\mathbb{R}^2)^*$$

where $\text{Sym}^m(\mathbb{R}^2)^*$ is the space of m -th order homogeneous polynomials (of dimension $m+1$). The action of V_4 on J_O^m is the composition of position permutation and the tensor power of the Jacobian:

$$(g \cdot Q)_i = Q_{g(i)} \circ (Jg)^{\otimes m}$$

where Jg is the Jacobian of g ($J_{\iota A} = \text{diag}(-1,1)$, $J_{\iota B} = \text{diag}(1,-1)$, $J_{\iota C} = \text{diag}(-1,-1)$), and $(Jg)^{\otimes m}$ denotes its natural action on the space of m -th order homogeneous polynomials (that is, $Q \circ Jg$).

By Theorem 9.4(a), a configuration of χ_{AB} type satisfies $g \cdot Q = \chi_{AB}(g)Q$, where $\chi_{AB}(\text{id}) = +1$, $\chi_{AB}(\iota_A) = -1$, $\chi_{AB}(\iota_B) = -1$, $\chi_{AB}(\iota_C) = +1$.

Since the regular representation $\mathbb{R}[V_4]$ of V_4 decomposes into the direct sum of four one dimensional irreducible representations $\chi_0 \oplus \chi_A \oplus \chi_B \oplus \chi_{AB}$, and $\text{Sym}^m(\mathbb{R}^2) \neq \{0\}$, the space J_O^m , as a V_4 module, contains the χ_{AB} irreducible component, with multiplicity $\dim \text{Sym}^m(\mathbb{R}^2) = m+1 \geq 2$.

Step 2 (explicit construction of a vector of χ_{AB} type).

Take the projection operator of χ_{AB} on the group algebra $\mathbb{R}[V_4]$:

$$P_{AB} = \frac{1}{4}(1 - \iota_A - \iota_B + \iota_C)$$

For any nonzero $Q_0 \in \text{Sym}^m(\mathbb{R}^2)$, define $Q^{(0)} = (Q_0, 0, 0, 0) \in J_O^m$ (supported only at the position s_0). Then:

$$P_{AB}(Q^{(0)}) = \frac{1}{4}(Q_0, -Q_0 \circ J_A, -Q_0 \circ J_B, +Q_0 \circ J_C)$$

where the signs of the components are determined by the projection operator and the position permutation rule: ι_A maps position 0 to position 1, hence $-(\iota_A \cdot Q^{(0)})_1 = -Q_{\iota_A(1)} \circ J_A = -Q_0 \circ J_A$; ι_B likewise gives $-Q_0 \circ J_B$; ι_C maps position 0 to position 3, hence $+(\iota_C \cdot Q^{(0)})_3 = +Q_{\iota_C(3)} \circ J_C = +Q_0 \circ J_C$.

By the fundamental property of the projection operator, $P_{AB}(Q^{(0)})$ automatically falls into the χ_{AB} irreducible subspace, that is:

$$\iota_A(P_{AB}(Q^{(0)})) = -P_{AB}(Q^{(0)}), \quad \iota_B(P_{AB}(Q^{(0)})) = -P_{AB}(Q^{(0)})$$

and since $Q_0 \neq 0$, the first component $\frac{1}{4}Q_0 \neq 0$; hence $PAB(Q^{(0)}) \neq 0$.

Step 3 (from the Jet vector to a smooth perturbation).

By the multivariate Borel lemma (or, more directly, by constructing a compactly supported smooth cut off function with the prescribed m-th order Taylor polynomial within each ball $B(si, r_0)$), for any $Q \in J_O^m$ there exists $\Phi \in C_c^\infty(\mathbb{R}^2)$ such that the m-th order Taylor expansion of Φ on the four element orbit $\mathcal{O}$ is exactly Q , and all Taylor coefficients of Φ on $\mathcal{O}$ of orders lower than m vanish. In particular, taking $Q = PAB(Q^{(0)})$ (a nonzero configuration of χ_{AB} type), there exists a nonzero smooth perturbation $\Phi AB \in \mathcal{D}$ (the smooth field space of Lemma 9.5C) whose m-th order Jet is of χ_{AB} type.

Step 4 (the negative energy computation).

Substitute ΦAB into the m-th order energy variation of the Mother Equation. By Lemma 9.5A, the eigenvalue of the energy matrix $\widetilde{K_{\text{geom}}}$ of the Mother Equation on the χ_{AB} irreducible subspace is $-C$ (a group algebra invariant, independent of the order m). After adding the regularization ϵI , its contribution on the m-th order Jet space is $+\epsilon \|Q\|_{\text{jet}}^2$, where $\|Q\|_{\text{jet}}^2 = \sum_i 0^3 \|Q_i\|_{F^2}^2 > 0$ is the standard Frobenius type norm on the m-th order homogeneous polynomials (induced by the standard inner product on $\mathbb{R}^2$). Hence:

$$\Delta E^{(m)}[\Phi AB] = (-C + \epsilon) \|Q\|_{\text{jet}}^2$$

From $Q \neq 0$ one has $\|Q\|_{\text{jet}}^2 > 0$.

By the intrinsic condition $C > \epsilon$ of Parts One to Eight:

$$\Delta E^{(m)}[\Phi AB] = -(C - \epsilon) \|Q\|_{\text{jet}}^2 < 0$$

strictly negative.

Step 5 (conclusion).

The unitarity axiom of the Mother Equation requires $\forall \delta \Phi \neq 0: \Delta E > 0$. But Steps 3 and 4 explicitly constructed a nonzero smooth perturbation ΦAB with $\Delta E^{(m)} < 0$, directly violating the axiom.

Therefore, whether m is odd or even, a nonreal m fold zero triggers the unitarity contradiction of the Mother Equation.

Conclusion of the contradiction self detonation.

As a dual channel unitary system, the Mother Equation grows the critical line LA on the complex plane through its geometric structure. Theorem 9.3A reduces the zero problem of doubly reflected entire functions to the zero distribution of the real entire function $g(w)$. Lemma 9.5A strictly established the lift of the delta kernel energy functional from the space of field values to the Jet space, preserving the group algebra eigenvalue. Theorem 9.5 derives: the unitarity axiom of the Mother Equation self consistently rejects any nonreal zero of $g(w)$.

By Steps 1 to 7 of Theorem 9.5, the Mother Equation has restricted the zeros of $g(w)$ to the real axis; Step 8 further uses the positivity of the Riemann ξ function on the real axis to compress the real zeros to the negative half axis $w \leq 0$. By Corollary 9.3B(i), $w \leq 0$ is strictly equivalent to $s = \frac{1}{2} \pm i\gamma$, that is, the critical line. Therefore, in the intersection of the two conditions, the unitarity axiom of the Mother Equation and the real axis positivity of the ξ function, the zeros can only lie on the critical line $Re(s) = 1/2$.

The physical contradiction self detonation has succeeded, the mathematical contradiction self detonation has succeeded, the nonreal zeros have no living space in the self consistency extension of the Mother Equation, and the

original Riemann Hypothesis is instantaneously established as a logical fact.

Volume Three: The Successful Forward Derivation of the Riemann Hypothesis

Part Ten: The Forward Unfolding of the Mother Equation as the Whole 1

10.1 The Definition of the Whole 1 and the Structural Self Growth

10.1.1 The Structural Definition of the Mother Equation (Not a Hypothesis)

Definition 10.1.1 (the Mother Equation as the Whole 1).

The Mother Equation is defined as an indivisible structural constraint whole, denoted the Whole 1. Its constituent elements are as follows:

(1) Entry condition: double reflection symmetry, that is, an entire function $f: \mathbb{C} \rightarrow \mathbb{C}$ satisfying

$$(A) f(s) = f(1-s), \forall s \in \mathbb{C}; (B) f(\bar{s}) = \overline{f(s)}, \forall s \in \mathbb{C};$$

(2) Repulsion mechanism: the unitarity axiom, that is, the energy response of the system to any nonzero perturbation is strictly positive,

$$\eta \equiv 1 \Leftrightarrow \forall \delta\Phi \neq 0: \Delta E[\delta\Phi] > 0;$$

(3) Geometric output: the reflection axes LA and LB forcibly grown from the dual channel kernel definition, and their unique intersection point.

Key remark: the Mother Equation here is not an equation to be solved, nor a set or function space containing particular solutions; it is defined as a Whole 1, a self consistency structure with global constraint power. Within this framework, "1" is not a number, but an indivisible structural constraint whole.

10.1.2 The Strict Definition of the Dual Channel Kernel

Definition 10.1.2 (the dual channel kernel structure).

The configuration space is defined as $X = \mathbb{C}$. The two kernel structure is defined as

$$\mathcal{K}(s, w) = \frac{1}{2}(\mathcal{K}_A + \mathcal{K}_B) + \varepsilon\delta(s - w),$$

where the channel A kernel $\mathcal{K}_A$ enforces the coupling of s with $\iota_A(s) = 1 - \bar{s}$:

$$\mathcal{K}_A(s, w) = C \quad (\text{if and only if}) \quad w = 1 - \bar{s};$$

and the channel B kernel $\mathcal{K}_B$ enforces the coupling of s with $\iota_B(s) = \bar{s}$:

$$\mathcal{K}_B(s, w) = C \quad (\text{if and only if}) \quad w = \bar{s}.$$

The coupling strength $C > 0$ is a classical geometric coupling parameter. The regularization parameter $\varepsilon \geq 0$ is the local fluctuation gap.

10.1.3 The Geometric Growth of the Reflection Axes (Element by Element Derivation)

Theorem 10.1.3 (the structural self growth of the reflection axes).

The dual channel kernel definition of the Mother Equation automatically grows two reflection axes on the complex plane:

Step 1: the fixed point set of channel A.

Solve the fixed point equation $s = \iota A(s) = 1 - \bar{s}$.

Writing $s = \sigma + i\gamma$, one has

$$\sigma + i\gamma = 1 - (\sigma - i\gamma) = 1 - \sigma + i\gamma$$

Comparing the real parts:

$$\sigma = 1 - \sigma \Rightarrow 2\sigma = 1 \Rightarrow \sigma = \frac{1}{2}.$$

The imaginary part is automatically satisfied: $\gamma = \gamma$.

Therefore the fixed point set of channel A is the critical line

$$L_A = \left\{ s \in \mathbb{C} : \text{Re}(s) = \frac{1}{2} \right\}$$

Step 2: the fixed point set of channel B.

Solve the fixed point equation $s = \iota B(s) = \bar{s}$.

Writing $s = \sigma + i\gamma$, one has

$$\sigma + i\gamma = \sigma - i\gamma \Rightarrow 2i\gamma = 0 \Rightarrow \gamma = 0.$$

Therefore the fixed point set of channel B is the real axis

$$L_B = \{ s \in \mathbb{C} : \text{Im}(s) = 0 \}$$

Step 3: the pure geometric fact of the unique intersection point.

Solve for the intersection $L_A \cap L_B$:

$$\text{Re}(s) = \frac{1}{2} \text{ and } \text{Im}(s) = 0 \Rightarrow s = \frac{1}{2}.$$

This number $\frac{1}{2}$ is not an external parameter, not the result of numerical fitting. It is the unique point grown by the dual channel definition of the Mother Equation in the geometry of the complex plane.

Key observation: the critical line $\text{Re}(s) = \frac{1}{2}$ is not "discovered" but "grown". It is the direct unfolding, in geometry, of the structural definition of the Mother Equation.

10.1.4 The Forward Protocol of Dynamically Structured Systems

Definition 10.1.4 (the forward protocol of dynamically structured systems).

Given a configuration space Ω , an intrinsic symmetry group $\mathcal{G}$, and an adjustment function $\mathcal{A}$ (unitarity / rigidity / viscosity constraints), the forward protocol of the Mother Equation is defined as the following five steps:

- (1) identify the orbit structure of $\mathcal{G}$ on Ω (the orbit stabilizer theorem);
- (2) construct the energy kernel $\mathcal{K}$ and compute its spectral decomposition over the group algebra;
- (3) traverse all orbit types and determine the signs of stability (positive definite / zero / negative definite);
- (4) extract the unique stable output manifold $\mathcal{M} \subset \Omega$;

(5) lock the legitimate configurations to lie in $\mathcal{M}$.

In the Riemann configuration: $\Omega = \mathbb{C}$, $\mathcal{G} = V_4 = \{\text{id}, \iota_A, \iota_B, \iota_C\}$, $\mathcal{A}$ is the unitarity axiom $\Delta E > 0$, and the unique stable output manifold is $\mathcal{M} = LA = \{s: \text{Re}(s) = \frac{1}{2}\}$.

10.2 The Complete Scan of the Energy Landscape and the Unique Stability Theorem

10.2.1 The Group Theoretic Structure of the Four Element Orbit

Definition 10.2.1 (orbit generation).

For any $s_0 = \sigma + i\gamma$ ($\gamma > 0$), the action of the two channels generates the closed set

$$\mathcal{O}(s_0) = \{s_0, s_1, s_2, s_3\} = \{\sigma + i\gamma, 1 - \sigma + i\gamma, \sigma - i\gamma, 1 - \sigma - i\gamma\}$$

Theorem 10.2.2 (the group theoretic root of the orbit dimension).

The group $V_4 = \{\text{id}, \iota_A, \iota_B, \iota_C\}$ acts on $s \in \mathbb{C}$; by the orbit stabilizer theorem:

$$|\mathcal{O}(s)| = \frac{|V_4|}{|\text{Stab}(s)|} = \frac{4}{|\text{Stab}(s)|}$$

If $s \notin L_A \cup L_B$: $\text{Stab}(s) = \{\text{id}\}$, $|\mathcal{O}(s)| = 4$ (four element expanded orbit).

If $s \in L_A$ but $s \notin L_B$ (that is, $\sigma = \frac{1}{2}, \gamma \neq 0$, $\text{Stab}(s) = \{\text{id}, \iota_A\}$, $|\mathcal{O}(s)| = 2$ (two element collapsed orbit).

If $s \in L_A \cap L_B$ (that is, $s = \frac{1}{2}$): $\text{Stab}(s) = V_4$, $|\mathcal{O}(s)| = 1$ (fixed point).

10.2.2 The Element by Element Construction of the Total Kernel Matrix

Case one: $\sigma \neq \frac{1}{2}$ (deviating from the critical line, four element expanded orbit).

Restrict the Mother Equation to $\mathcal{O}(s_0) = \{s_0, s_1, s_2, s_3\}$ and define the field values $\Phi_i = \Phi(s_i)$, $\mathcal{K}ij = \mathcal{K}(s_i, s_j)$.

The element by element computation of the channel A kernel $\mathcal{K}A$ (all 16 matrix elements verified):

(i,j)	compute $1 - \bar{s}i$	equals s_j ?	result
(0,0)	$1 - (\sigma - i\gamma) = 1 - \sigma + i\gamma$	$\neq \sigma + i\gamma$ (since $\sigma \neq \frac{1}{2}$)	0
(0,1)	$1 - \sigma + i\gamma$	$= s_1$	C
(0,2)	$1 - \sigma + i\gamma$	$\neq \sigma - i\gamma$	0
(0,3)	$1 - \sigma + i\gamma$	$\neq 1 - \sigma - i\gamma$ (since $\gamma \neq 0$)	0
(1,0)	$1 - (1 - \sigma - i\gamma) = \sigma + i\gamma$	$= s_0$	C
(1,1)	$\sigma + i\gamma$	$\neq 1 - \sigma + i\gamma$	0
(1,2)	$\sigma + i\gamma$	$\neq \sigma - i\gamma$	0
(1,3)	$\sigma + i\gamma$	$\neq 1 - \sigma - i\gamma$	0
(2,0)	$1 - (\sigma + i\gamma) = 1 - \sigma - i\gamma$	$\neq \sigma + i\gamma$	0
(2,1)	$1 - \sigma - i\gamma$	$\neq 1 - \sigma + i\gamma$	0

(2,2)	$1 - \sigma - i\gamma$	$\neq \sigma - i\gamma$	0
(2,3)	$1 - \sigma - i\gamma$	$= s_3$	C
(3,0)	$1 - (1 - \sigma + i\gamma) = \sigma - i\gamma$	$\neq \sigma + i\gamma$	0
(3,1)	$\sigma - i\gamma$	$\neq 1 - \sigma + i\gamma$	0
(3,2)	$\sigma - i\gamma$	$= s_2$	C
(3,3)	$\sigma - i\gamma$	$\neq 1 - \sigma - i\gamma$	0

Therefore

$$\mathcal{K}A = C \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

The element by element computation of the channel B kernel $\mathcal{K}B$ (all 16 matrix elements verified):

(i,j)	compute $\bar{s}_i$	equals s_j ?	result
(0,0)	$\sigma - i\gamma$	$\neq \sigma + i\gamma$	0
(0,1)	$\sigma - i\gamma$	$\neq 1 - \sigma + i\gamma$	0
(0,2)	$\sigma - i\gamma$	$= s_2$	C
(0,3)	$\sigma - i\gamma$	$\neq 1 - \sigma - i\gamma$	0
(1,0)	$1 - \sigma - i\gamma$	$\neq \sigma + i\gamma$	0
(1,1)	$1 - \sigma - i\gamma$	$\neq 1 - \sigma + i\gamma$	0
(1,2)	$1 - \sigma - i\gamma$	$\neq \sigma - i\gamma$	0
(1,3)	$1 - \sigma - i\gamma$	$= s_3$	C
(2,0)	$\sigma + i\gamma$	$= s_0$	C
(2,1)	$\sigma + i\gamma$	$\neq 1 - \sigma + i\gamma$	0
(2,2)	$\sigma + i\gamma$	$\neq \sigma - i\gamma$	0
(2,3)	$\sigma + i\gamma$	$\neq 1 - \sigma - i\gamma$	0
(3,0)	$1 - \sigma + i\gamma$	$\neq \sigma + i\gamma$	0
(3,1)	$1 - \sigma + i\gamma$	$= s_1$	C
(3,2)	$1 - \sigma + i\gamma$	$\neq \sigma - i\gamma$	0
(3,3)	$1 - \sigma + i\gamma$	$\neq 1 - \sigma - i\gamma$	0

Therefore

$$\mathcal{K}B = C \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

The geometric kernel:

$$\mathcal{K}_{\text{geom}} = \frac{1}{2}(\mathcal{K}_A + \mathcal{K}_B) = \frac{C}{2} \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}$$

The total kernel:

$$\mathcal{K} = \mathcal{K}_{\text{geom}} + \varepsilon I = \frac{C}{2} \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix} + \varepsilon \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

10.2.3 The Strict Computation of the Eigenvalues

Step 1: the eigenvalues of the geometric kernel $\mathcal{K}_{\text{geom}}$.

Let M be the coefficient matrix of $\mathcal{K}_{\text{geom}}$ divided by $C/2$:

$$M = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}$$

Observation: $\text{row}0 = \text{row}3 = [0,1,1,0]; \text{row}1 = \text{row}2 = [1,0,0,1]$

Verifying the four eigenvectors:

(1) $v_0 = (1,1,1,1)$ (the trivial representation χ_0):

$$Mv_0 = (2,2,2,2) = 2v_0 \Rightarrow \lambda = 2 \Rightarrow \text{eigenvalue of } \mathcal{K}_{\text{geom}} = C$$

(2) $v_A = (1,-1,1,-1)$ (the representation χ_A):

$$Mv_A = (0,0,0,0) = 0 \Rightarrow \lambda = 0 \Rightarrow \text{eigenvalue of } \mathcal{K}_{\text{geom}} = 0$$

(3) $v_B = (1,1,-1,-1)$ (the representation χ_B):

$$Mv_B = (0,0,0,0) = 0 \Rightarrow \lambda = 0 \Rightarrow \text{eigenvalue of } \mathcal{K}_{\text{geom}} = 0$$

(4) $v_{AB} = (1,-1,-1,1)$ (the representation χ_{AB}):

$$Mv_{AB} = (-2,2,2,-2) = -2v_{AB} \Rightarrow \lambda = -2 \Rightarrow \text{eigenvalue of } \mathcal{K}_{\text{geom}} = -C$$

Step 2 (hand expansion): the strict expansion of the characteristic polynomial.

$$\det(M - \lambda I) = \det \begin{pmatrix} -\lambda & 1 & 1 & 0 \\ 1 & -\lambda & 0 & 1 \\ 1 & 0 & -\lambda & 1 \\ 0 & 1 & 1 & -\lambda \end{pmatrix}$$

Expanding along the first row:

$$= -\lambda \cdot \det \begin{pmatrix} 0 & -\lambda & 1 \\ 1 & 1 & -\lambda \end{pmatrix} - 1 \cdot \det \begin{pmatrix} 1 & 0 & 1 \\ 1 & -\lambda & 1 \end{pmatrix} + 1 \cdot \det \begin{pmatrix} 1 & -\lambda & 1 \\ 1 & 0 & 1 \end{pmatrix}$$

First term:

$$-\lambda \left[-\lambda \cdot \det \begin{pmatrix} -\lambda & 1 \\ 1 & -\lambda \end{pmatrix} - 0 + 1 \cdot \det \begin{pmatrix} 0 & -\lambda \\ 1 & 1 \end{pmatrix} \right] = -\lambda [-\lambda(\lambda^2 - 1) + (0 + \lambda)] = -\lambda(-\lambda^3 + 2\lambda) = \lambda^4 - 2\lambda^2$$

Second term:

$$-1 \left[1 \cdot \det \begin{pmatrix} -\lambda & 1 \\ 1 & -\lambda \end{pmatrix} + 1 \cdot \det \begin{pmatrix} 1 & -\lambda \\ 0 & 1 \end{pmatrix} \right] = -1[(\lambda^2 - 1) + 1] = -\lambda^2$$

Third term:

$$+1 \left[1 \cdot \det \begin{pmatrix} 0 & 1 \\ 1 & -\lambda \end{pmatrix} - (-\lambda) \cdot \det \begin{pmatrix} 1 & 1 \\ 0 & -\lambda \end{pmatrix} + 1 \cdot \det \begin{pmatrix} 1 & -\lambda \\ 0 & 1 \end{pmatrix} \right] = 1[(0 - 1) + \lambda(-\lambda) + 1] = -\lambda^2$$

Sum:

$$\lambda^4 - 2\lambda^2 - \lambda^2 - \lambda^2 = \lambda^4 - 4\lambda^2 = \lambda^2(\lambda^2 - 4) = \lambda^2(\lambda - 2)(\lambda + 2)$$

Hence the eigenvalues are $\lambda \in \{0, 0, 2, -2\}$, corresponding to the eigenvalues $\{0, 0, C, -C\}$ of $\mathcal{K}_{\text{geom}}$, in complete agreement with the verification of the four eigenvectors.

Step 3: the eigenvalues of the total kernel $\mathcal{K} = \mathcal{K}_{\text{geom}} + \varepsilon I$.

After adding the regularization, the eigenvalues become

$$\{C + \varepsilon, \varepsilon, \varepsilon, -C + \varepsilon\}$$

10.2.4 The Strict Proof of the Phase Transition Theorem

Theorem 10.2.4 (the phase transition theorem).

Under the definition of the Mother Equation as a dual channel unitary system, the energy landscape on the complex plane has a strict phase transition structure:

1. When $\sigma \neq \frac{1}{2}$ (deviating from the critical line): the four element orbit remains four dimensionally expanded, and the χ_{AB} mode contributes the strictly negative eigenvalue

$$\lambda_{\chi_{AB}} = -C + \varepsilon$$

By the intrinsic boundary condition $C > \varepsilon \geq 0$ of the Mother Equation, one has $\lambda_{\chi_{AB}} < 0$. The system is in a structurally unstable state.

2. When $\sigma = \frac{1}{2}$ (on the critical line):

the four element orbit collapses into a two element orbit. The discretization then automatically degenerates to the number of independent physical points, avoiding double counting.

On the collapsed subspace, $s_0 = s_1$ and $s_2 = s_3$, and the independent points are $\{s_0, s_2\}$. The total kernel restricted to this subspace:

$$\mathcal{K}_{\text{geom}}|_{\text{sub}} = \frac{c}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \mathcal{K}|_{\text{sub}} = \frac{c}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \varepsilon \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Step 2.1 (hand expansion): the strict computation of the eigenvalues on the collapsed subspace.

$$\det \left(\frac{c}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} - \lambda I \right) = \det \begin{pmatrix} \frac{c}{2} - \lambda & \frac{c}{2} \\ \frac{c}{2} & \frac{c}{2} - \lambda \end{pmatrix} = \left(\frac{c}{2} - \lambda \right)^2 - \left(\frac{c}{2} \right)^2 = 0$$

$$\Rightarrow \left(\frac{C}{2} - \lambda\right)^2 = \left(\frac{C}{2}\right)^2 \Rightarrow \frac{C}{2} - \lambda = \pm \frac{C}{2}$$

Hence $\lambda_1 = \frac{C}{2} + \frac{C}{2} = C$ (the symmetric mode (1, 1)) and $\lambda_2 = \frac{C}{2} - \frac{C}{2} = 0$ (the antisymmetric mode (1, -1)). After adding the regularization εI : $\lambda_1 = C + \varepsilon > 0$. Both are strictly positive. The system is in a strictly stable state.

The strict statement of the absence of intermediate states.

By the orbit stabilizer theorem, the order $|\text{Stab}(s)|$ of the stabilizer can only take the values 1, 2, 4, corresponding to the orbit dimensions 4, 2, 1. No other possibility exists between these three integers. Therefore, when σ deviates from $\frac{1}{2}$ by an arbitrarily small amount $\delta \neq 0$, the orbit dimension jumps from 2 to 4, and the energy of the χ_{AB} mode jumps from $\varepsilon > 0$ to $-C + \varepsilon < 0$. This phase transition is discrete and irreversible; there is no asymptotic transition zone.

Proof: already strictly verified by the matrix computations above.

10.2.5 The Unique Stable Output Theorem

Theorem 10.2.5 (the unique stable output theorem).

Under the definition of the Mother Equation as a dual channel unitary system, the energy landscape on the complex plane has a unique stable output manifold, namely the critical line

$$L_A = \left\{s \in \mathbb{C} : \text{Re}(s) = \frac{1}{2}\right\}$$

For any $\gamma > 0$, the minimal eigenvalue at the point $\frac{1}{2} + i\gamma$ is $\varepsilon > 0$; for $\gamma = 0$, the fixed point $s = \frac{1}{2}$ is the global energy minimum.

Note (numerical consistency between the upper and lower volumes): the minimal eigenvalue $\varepsilon > 0$ on the collapsed subspace here (the energy of the antisymmetric mode (1, -1) on the two element orbit) is equivalent, in its qualitative conclusion of strict positive definiteness, to the "collapsed orbit χ_{AB} mode energy $2\varepsilon > 0$ " in the table of Section 8.3 of the upper volume. The quantitative difference comes only from the normalization of the subspace restriction: the upper volume states the total orbital energy (each of the two points contributing ε), while the lower volume states the strict eigenvalues of the 2×2 matrix. Both confirm the strict stability on the critical line.

Proof:

Step 1: For any $\gamma > 0$, when $\sigma = \frac{1}{2}$, by Theorem 10.2.4(2), the minimal eigenvalue is $\lambda_{\min} = \varepsilon > 0$ on the two element subspace, the two eigenvalues are $C + \varepsilon$ and ε , the minimum being ε). Note: this strict positive definiteness of $\varepsilon > 0$ is guaranteed by the intrinsic boundary $C > \varepsilon \geq 0$ of the Mother Equation, and is independent of the concrete value of C. Therefore all points on the critical line are stable.

Step 2: For any $\sigma \neq \frac{1}{2}$, whatever value $\gamma > 0$ takes, by Theorem 10.2.4(1), the minimal eigenvalue is $\lambda_{\min} = -C + \varepsilon < 0$ herefore all points deviating from the critical line are unstable.

Step 3: There are no intermediate states. By the strict statement above, the order $|\text{Stab}(s)| \in \{1, 2, 4\}$ is a discrete set of integers, and there is no other possibility among the orbit dimensions 4, 2, 1. When σ deviates from $\frac{1}{2}$ by an arbitrarily small amount δ , the orbit dimension jumps from 2 to 4, and the energy of the χ_{AB} mode jumps from $\varepsilon > 0$ to $-C + \varepsilon < 0$.

Conclusion: L_A is the unique stable output manifold of the Mother Equation system on the entire complex plane.

10.3 The Stable Region and the Legitimacy of Field Configurations

10.3.1 The Strict Definition of Legitimate Configurations

Definition 10.3.1 (legitimate configuration).

Within the framework of the Whole 1 of the Mother Equation, a field configuration Φ is called a legitimate configuration if and only if its support lies entirely within the stable spectrum of the Mother Equation. That is, for the support point $s_0 = \sigma_0 + i\gamma_0$ ($\gamma_0 > 0$) of a perturbation $\delta\Phi$, one has

$$\lambda \min(\sigma_0, \gamma_0) > 0$$

By Theorem 10.2.5, this condition is equivalent to

$$\sigma_0 = \frac{1}{2}$$

10.3.2 The Structural Determination of Nonreal Deviating Configurations

Theorem 10.3.2 (the illegitimacy of nonreal deviating configurations).

Any field configuration satisfying $\operatorname{Re}(s) \neq \frac{1}{2}$ and $\operatorname{Im}(s) \neq 0$ is not a legitimate configuration of the Mother Equation.

Proof: Let $s_0 = \sigma_0 + i\gamma_0$ satisfy $\sigma_0 \neq \frac{1}{2}$ and $\gamma_0 > 0$. By Theorem 10.2.5, this point lies in the unstable region of the energy landscape, $\lambda \min(\sigma_0, \gamma_0) = -C + \varepsilon < 0$. Therefore the field configuration at this point violates the unitarity axiom $\forall \delta\Phi \neq 0: \Delta E[\delta\Phi] > 0$. By Definition 10.3.1, this configuration is not legitimate.

Key epistemological turn: what happens here is not the reverse logic of "assume a nonreal configuration exists and then derive a contradiction", but a direct determination: nonreal deviating configurations have no legitimate place of existence in the stable spectrum of the Mother Equation. This is not repulsion, but the direct presentation of the extension of structural self consistency: the definitional level of the Mother Equation naturally contains no living space for such configurations.

Part Eleven: From Structural Growth to the Locking of the Zeros

11.1 The Structural Dimension Reduction of Doubly Reflected Entire Functions

11.1.1 The Complete Proof of Theorem 9.3A

Theorem 11.1.1 (the structural dimension reduction of doubly reflected entire functions).

For any $f \in \mathcal{F}$ (the space of doubly reflected entire functions), f admits a unique representation

$$f(s) = g\left(\left(s - \frac{1}{2}\right)^2\right)$$

where $g(w)$ is an entire function with real coefficients.

Proof (step by step, strict):

Step 1: Taylor expansion at $s = \frac{1}{2}$:

$$f(s) = \sum_{n=0}^{\infty} c_n \left(s - \frac{1}{2}\right)^n, \quad c_n = \frac{f^{(n)}\left(\frac{1}{2}\right)}{n!}$$

Step 2: By condition (A), $f(s) = f(1-s)$. Letting $w = s - \frac{1}{2}$, one has $1-s = \frac{1}{2} - w$;

Hence $f\left(\frac{1}{2} + w\right) = f\left(\frac{1}{2} - w\right)$, that is,

$$\sum n = 0^\infty c_n w^n = \sum n = 0^\infty c_n (-w)^n = \sum n = 0^\infty c_n (-1)^n w^n$$

Comparing coefficients of equal powers: $c_n = (-1)^n c_n$. When n is odd, $c_n = -c_n \Rightarrow c_n = 0$.

Therefore all odd order coefficients vanish, and f is even about $\frac{1}{2}$:

$$f(s) = \sum k = 0^\infty c_{2k} \left(s - \frac{1}{2}\right)^{2k}$$

Step 3: Let $z = \left(s - \frac{1}{2}\right)^2$ and define

$$g(z) = \sum k = 0^\infty c_{2k} z^k$$

Then $f(s) = g\left(\left(s - \frac{1}{2}\right)^2\right)$. Since f is entire, g is also entire (analyticity of the composition: $z \mapsto \frac{1}{2} + \sqrt{z}$ is analytic on $\mathbb{C} \setminus (-\infty, 0]$, and the evenness of f guarantees that g is continuous across the branch cut, so that g admits an analytic continuation to an entire function).

Step 4: Deriving the coefficients $a_k = c_{2k} \in \mathbb{R}$.

By condition (B), $f(\overline{s}) = \overline{f(s)}$. Taking $s = \frac{1}{2} + t$ ($t \in \mathbb{R}$), one has

$$f\left(\frac{1}{2} + t\right) = \overline{f\left(\frac{1}{2} + t\right)} \in \mathbb{R}$$

so f takes real values on the real axis. In particular, $f\left(\frac{1}{2}\right) = g(0) = a_0 \in \mathbb{R}$.

For $g(w) = \sum a_k w^k$, taking $w \geq 0$ (a real number), one has

$$g(w) = f\left(\frac{1}{2} + \sqrt{w}\right) \in \mathbb{R}$$

The function g is entire and takes real values on the interval $[0, +\infty)$. By the identity theorem for analytic functions, if a power series takes real values on a real interval, all its coefficients must be real. Hence $a_k \in \mathbb{R}^{[5]}$.

Step 5: Uniqueness.

If $f(s) = g_1\left(\left(s - \frac{1}{2}\right)^2\right) = g_2\left(\left(s - \frac{1}{2}\right)^2\right)$, then $g_1(z) = g_2(z)$ holds for infinitely many z (in fact for all $z \in \mathbb{C}$).

By the identity theorem, $g_1 = g_2^{[5]}$.

Forward reading: this dimension reduction is not a simplifying trick, but a structural necessity forcibly imposed by double reflection symmetry. Once one enters the soil of this symmetry, entire functions must grow in the form $g\left(\left(s - \frac{1}{2}\right)^2\right)$; no other form is possible. The change of variables $w = \left(s - \frac{1}{2}\right)^2$ is the natural folding of the structure of the Mother Equation on the complex plane.

11.1.2 The Structured Presentation of the Zero Classification

Corollary 11.1.2 (classification of the zeros).

For any zero w_0 of $g(w)$, let $z_0 = \sqrt{w_0}$ (principal branch). Then the zeros of $f(s)$ are $s = \frac{1}{2} \pm z_0$.

Case (i): $w_0 \in (-\infty, 0)$ (negative real).

In this case $z_0 = \pm i\gamma$ (purely imaginary, $\gamma = \sqrt{|w_0|} > 0$), hence

$$s = \frac{1}{2} \pm i\gamma$$

and the zero lies on the critical line $\text{Re}(s) = \frac{1}{2}$.

Case (ii): $w_0 \in (0, +\infty)$ (positive real).

In this case $z_0 = \pm\gamma$ (real, $\gamma = \sqrt{w_0} > 0$), hence

$$s = \frac{1}{2} \pm \gamma$$

and the zero lies on the real axis.

Case (iii): $w_0 \notin \mathbb{R}$ (nonreal).

In this case $z_0 = \delta + i\gamma$ ($\delta \neq 0, \gamma \neq 0$), hence

$$s = \frac{1}{2} \pm (\delta + i\gamma) = \frac{1}{2} \pm \delta \pm i\gamma,$$

and the zero deviates from the critical line ($\text{Re}(s) = \frac{1}{2} \pm \delta \neq \frac{1}{2}$).

Key equivalence:

the Riemann Hypothesis $\Leftrightarrow$ all zeros of $g(w)$ are negative real numbers $w \leq 0$

11.2 The Forward Locking of the Zero Positions

11.2.1 The Legitimacy Constraint on Zero Configurations

Theorem 11.2.1 (the forward constraint on zero configurations).

Let $f \in \mathcal{F}$, and let f , as a field configuration of the Mother Equation, satisfy the unitarity axiom (that is, f is a legitimate configuration). Then the zeros of f can only lie in the stable region of the energy landscape of the Mother Equation.

Explicit account of the association: concretely, for $f \in \mathcal{F}$, let $v = \text{Im}(f)$. By Theorem 9.4 of the middle volume, v is a real valued field of χ_{AB} type, and at a zero ρ one has $v(\rho) = 0$ while $\nabla v(\rho) \neq 0$ (the simple zero case). Therefore v can be regarded as a trial field configuration under the framework of the Mother Equation, and its zero support point ρ must obey the stability constraints of the Mother Equation.

Proof: For $f \in \mathcal{F}$, its imaginary part $v = \text{Im}(f)$ constitutes a natural trial field configuration under the framework of the Mother Equation (by Theorem 9.4 of the middle volume, v is a real valued field of χ_{AB} type). The zero support set of this field configuration must lie within the stable spectrum of the Mother Equation. By Definition 10.3.1, a legitimate configuration must lie within the stable spectrum. By Theorem 10.2.5, the unique stable region is the intersection structure of LA (the critical line) and LB (the real axis). Therefore the zero configuration must obey this structural constraint.

11.2.2 The Structural Exclusion of Nonreal Zeros

(complete proof, including the Jet lift bridge and the hand computation of the partial derivatives)

Theorem 11.2.2 (the structural exclusion of nonreal zeros).

$g(w)$ cannot have nonreal zeros.

Forward proof:

Step 1: Suppose g has a nonreal zero $w_0 = u + iv$ ($v \neq 0$). Let $z_0 = \sqrt{w_0} = \delta + i\gamma$ (principal branch, $\delta \neq 0$, $\gamma \neq 0$).

Step 2: $f(s) = g((s - \frac{1}{2})^2)$ vanishes at the following four points:

$$s_0 = \frac{1}{2} + \delta + i\gamma$$

$$s_1 = \frac{1}{2} - \delta + i\gamma \quad (= \iota_A(s_0))$$

$$s_2 = \frac{1}{2} + \delta - i\gamma \quad (= \iota_B(s_0))$$

$$s_3 = \frac{1}{2} - \delta - i\gamma \quad (= \iota_C(s_0))$$

The four points are pairwise distinct (since $\delta \neq 0$, $\gamma \neq 0$) and form the four element orbit of V_4 .

Step 3: By Theorem 10.2.5, this quadruple satisfies $\operatorname{Re}(s_0) \neq \frac{1}{2}$ and $\operatorname{Im}(s_0) \neq 0$, that is, it lies in the unstable region of the energy landscape of the Mother Equation. The energy of the χ_{AB} mode in this region is strictly negative ($\lambda\chi_{AB} = -C + \varepsilon < 0$).

Step 3 (continued): the Jet lift bridge and the hand computation of the partial derivatives. By Theorem 9.4 of the middle volume (the χ_{AB} structure of the local germ space), if ρ is a simple zero of f and $\sigma \neq \frac{1}{2}$, then $\nabla v(\rho) \neq 0$ (where $v = \operatorname{Im}(f)$). The following gives the strict derivation of the gradient vector $\mathbf{w}$ component by component:

Let $v(x, y) = \operatorname{Im}(f(x + iy))$. From the double reflection symmetry:

(a) the partial derivative relation under the action of ι_A : $v(1 - x, y) = -v(x, y)$.

Differentiating in x (chain rule):

$$\frac{\partial}{\partial x} v(1 - x, y) = \partial_1 v(1 - x, y) \cdot (-1) = -\partial x' v(1 - x, y)$$

where $x' = 1 - x$. From $v(1 - x, y) = -v(x, y)$, differentiating both sides in x :

$$-\partial x' v(1 - x, y) = -\partial_x v(x, y) \Rightarrow \partial_x v(1 - x, y) = \partial_x v(x, y)$$

Hence at $s_1 = (1 - \sigma, \gamma) = \iota_A(s_0)$:

$$\partial_x v(s_1) = \partial_x v(s_0) = a$$

Differentiating in y :

$$\frac{\partial}{\partial y} v(1-x, y) = \partial_2 v(1-x, y) \cdot 1 = \partial_y v(1-x, y)$$

From $v(1-x, y) = -v(x, y)$, differentiating both sides in y :

$$\partial_y v(1-x, y) = -\partial_y v(x, y)$$

Hence at s_1 :

$$\partial_y v(s_1) = -\partial_y v(s_0) = -b$$

(b) the partial derivative relation under the action of ι_B : $v(x, -y) = -v(x, y)$.

Differentiating in x :

$$\frac{\partial}{\partial x} v(x, -y) = \partial_1 v(x, -y) \cdot 1 = \partial_x v(x, -y)$$

From $v(x, -y) = -v(x, y)$, differentiating both sides in x :

$$\partial_x v(x, -y) = -\partial_x v(x, y)$$

Hence at $s_2 = (\sigma, -\gamma) = \iota_B(s_0)$:

$$\partial_x v(s_2) = -\partial_x v(s_0) = -a$$

Differentiating in y (chain rule):

$$\frac{\partial}{\partial y} v(x, -y) = \partial_2 v(x, -y) \cdot (-1) = -\partial_y v(x, -y)$$

From $v(x, -y) = -v(x, y)$, differentiating both sides in y :

$$-\partial_y v(x, -y) = -\partial_y v(x, y) \Rightarrow \partial_y v(x, -y) = \partial_y v(x, y)$$

Hence at s_2 :

$$\partial_y v(s_2) = \partial_y v(s_0) = b$$

(c) The partial derivatives at $s_3 = (1-\sigma, -\gamma) = \iota_A(\iota_B(s_0))$:

Applying ι_A to s_2 :

$$\partial_x v(s_3) = \partial_x v(s_2) = -a$$

$$\partial_y v(s_3) = -\partial_y v(s_2) = -b$$

In summary, the eight dimensional gradient vector:

$$\mathbf{w} = (\partial_x v(s_0), \partial_y v(s_0), \partial_x v(s_1), \partial_y v(s_1), \partial_x v(s_2), \partial_y v(s_2), \partial_x v(s_3), \partial_y v(s_3)) = (a, b, a, -b, -a, b, -a, -b)$$

This result agrees completely with the strict verification of 9.4 in the middle volume.

By Lemma 9.5A (middle volume), the first order Jet expansion of the energy functional of the Mother Equation at the zeros is the quadratic form:

$$\Delta E^{(1)} = \langle \mathbf{w}, \tilde{\mathcal{K}} \mathbf{w} \rangle = (-C + \varepsilon) \|\mathbf{w}\|^2 < 0$$

Therefore, a nonreal zero not only deviates geometrically from the critical line, but also activates a strictly negative energy mode in the energy landscape of the Mother Equation. This negative energy is a group algebra invariant, independent of regularization operations or dimensional lifts.

Step 4: By Theorem 11.2.1, the zero configuration, as a subset of legitimate field configurations, must lie within the stable region. But Step 3 shows that the four element orbit induced by a nonreal zero lies in the unstable region, and its gradient Jet vector produces a strictly negative energy.

Step 5: Therefore, the zero configuration induced by a nonreal zero w_0 has no legitimate place of existence in the stable spectrum of the Mother Equation.

Step 5 (continued): By Corollary 11.1.2, the zero forms exhaust three types (critical line / real axis / nonreal deviation). Type (iii) (nonreal deviation) has been proved to have no legitimate place in the stable spectrum of the Mother Equation. Therefore the zeros of $g(w)$ can only be of type (i) or type (ii), that is, real numbers.

Step 6: All zeros of $g(w)$ must be real (nonpositive or negative).

Note: this proof does not rely on the reverse logic of "assuming a counterexample exists"; it determines directly, through the structural constraint of the stable spectrum of the Mother Equation, that nonreal zero configurations do not lie in the legitimate solution space.

11.2.3 Generalization to Multiple Zeros

Note 11.2.3 (multiple zeros).

For any m fold zero ($m \geq 1$), the conclusion of Theorem 11.2.2 still holds.

Derivation: The unitarity axiom of the Mother Equation requires $\Delta E > 0$ for all nonzero perturbations $\delta\Phi \equiv 0$. Therefore, it suffices to construct any single nonzero perturbation with $\Delta E < 0$ to violate the axiom.

For an m fold zero, consider the direct sum of the m-th order symmetric tensors over the four element orbit:

$$J_0^m = \bigoplus_{i=0}^3 \text{Sym}^m(\mathbb{R}^2)^*$$

The action of V_4 on this space contains the χ_{AB} irreducible component, with multiplicity $\dim \text{Sym}^m(\mathbb{R}^2) = m + 1 \geq 2$. Construct the projection operator:

$$PAB = \frac{1}{4}(1 - \iota_A - \iota_B + \iota_C)$$

For any nonzero $Q_0 \in \text{Sym}^m(\mathbb{R}^2)$, $PAB(Q^{(0)})$ yields a nonzero vector of χ_{AB} type. The eigenvalue of the energy matrix of the Mother Equation on the χ_{AB} subspace is still $-C$ (a group algebra invariant, independent of the order m). After adding the regularization:

$$\Delta E^{(m)}[\Phi_{AB}] = (-C + \varepsilon)\|Q\|\text{jet}^2 < 0$$

Therefore, whether m is odd or even, a nonreal m fold zero triggers the unitarity contradiction of the Mother Equation.

11.3 Real Axis Positivity and the Locking of the Critical Line

Lemma 11.3 (real axis positivity exclusion).

Step one (number theoretic input): completing the argument for real axis positivity and the derivation of $g(w)$.

Combining known classical facts, the Riemann ξ function is strictly positive on the real interval $(0, 1)$ [2].

Strict argument:

For $s \in (0,1)$: the factor $s(s-1) < 0$ (since $s > 0$, $s-1 < 0$), while $\zeta(s) < 0$ (by the functional equation of ζ and its analytic properties on $(0, 1)$; $\zeta(0) = -\frac{1}{2}$, and ζ has no zeros on $(0, 1)$ and keeps a negative sign); hence the product $s(s-1)\zeta(s) > 0$ (negative times negative is positive). The remaining factors satisfy $\pi^{-s/2}\Gamma(s/2) > 0$. Therefore

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s) > 0$$

For $s > 1$: the series $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$ has positive terms, hence $\zeta(s) > 0$, and all factors are obviously positive; therefore $\xi(s) > 0$.

By $\xi(s) = \xi(1-s)$, the real axis positivity covers the whole of $s \in \mathbb{R} \setminus \{0,1\}$.

Step two (compression to the negative half axis): complete derivation. The real zeros of $g(w)$ can only satisfy $w \leq 0$ (negative real numbers).

From $f(s) = g\left(\left(s-\frac{1}{2}\right)^2\right)$ and the strict positivity of $\xi(s)$:

For $s \in (0,1)$, let $t = s - \frac{1}{2} \in (-\frac{1}{2}, \frac{1}{2})$; then $w = t^2 \in [0, \frac{1}{4})$. From $\xi(s) = g((s-\frac{1}{2})^2) = g(t^2)$ and $\xi(s) > 0$, one has

$$g(w) > 0 \quad \text{for } w \in \left[0, \frac{1}{4}\right)$$

For $s > 1$, from $\xi(s) > 0$ and $s = \frac{1}{2} + \sqrt{w}$ ($w > \frac{1}{4}$), one has

$$g(w) = \xi\left(\frac{1}{2} + \sqrt{w}\right) > 0 \quad \text{for } w > \frac{1}{4}$$

For $s < 0$, from $\xi(s) = \xi(1-s)$ and $1-s > 1$, one has $\xi(s) = \xi(1-s) > 0$; hence

$$g(w) = \xi\left(\frac{1}{2} + \sqrt{w}\right) = \xi\left(\frac{1}{2} - \sqrt{w}\right) > 0 \quad \text{for } w > \frac{1}{4}$$

(here $\frac{1}{2} - \sqrt{w} < 0$, corresponding to the case $s < 0$, whose positivity is guaranteed by the symmetry $\xi(s) = \xi(1-s)$).

In summary:

$$g(w) > 0 \text{ for all } w \geq 0$$

Therefore the real zeros of g (if any) must satisfy $w < 0$, that is, lie on the negative half axis.

Step three (locking of the critical line): By Corollary 11.1.2(i):

$$w \leq 0 \Leftrightarrow s = \frac{1}{2} \pm i\sqrt{|w|}$$

that is, the zeros lie strictly on the critical line $\text{Re}(s) = 1/2$.

Part Twelve: The Formation of the Bidirectional Closure under the Unity of the Three Volumes, and the Instantaneous Establishment of the Riemann Hypothesis

12.1 The Result of the Forward Derivation

Theorem 12.1 (the forward growth theorem of the Riemann Hypothesis).

Under the constraint of the dual channel unitary structure of the Whole 1 of the Mother Equation, the zero distribution of doubly reflected entire functions, as the direct unfolding of the extension of structural self consistency, lies strictly on the critical line $\text{Re}(s) = 1/2$.

Proof: By Theorem 10.2.5 (the unique stable output theorem), the unique stable output manifold of the Mother Equation is the critical line; by Theorem 11.2.1 (the forward constraint on zero configurations), legitimate zero configurations must lie in the stable region; by Theorem 11.2.2 (the structural exclusion of nonreal zeros), the nonreal zeros are excluded; by Lemma 11.3 (the real axis positivity exclusion), the positive real axis zeros are excluded. Therefore the zeros can only lie on the critical line.

Epistemological position: the success of the forward derivation of the Riemann Hypothesis is not a "theorem that has been proved", but a logical fact instantaneously presented upon the completion of the structural definition of the Mother Equation. That is, it is not a conclusion awaiting audit, but a rigid attribute exposed simultaneously with the unfolding of the self consistency of the Whole 1.

12.2 A Summary of the Unity of the Three Volumes

Theorem 12.2 (the bidirectional equivalence theorem). Within the framework of the Whole 1 of the Mother Equation, the following two propositions are logically equivalent:

(A) the self consistency extension of the Mother Equation contains no living space for nonreal zeros (reverse repulsion);

(B) the unique stable output manifold of the energy landscape of the Mother Equation on the complex plane is the critical line (forward growth).

The two together are equivalent to: the zeros of doubly reflected entire functions can only lie on $\text{Re}(s) = \frac{1}{2}$.

The equivalence of the upper, middle, and lower volumes.

The upper, middle, and lower volumes of this paper share the same mathematical core and structural constraints, but their epistemological postures complement one another:

Upper volume (physical self detonation): displayed in the language of physical energy, configurations deviating from the critical line produce strictly negative energy and are repelled by the unitarity axiom.

Middle volume (mathematical self detonation): displayed in the language of mathematical analysis, nonreal zeros activate the negative eigenvalue of the χ_{AB} mode and are repelled by structural self consistency.

Lower volume (forward growth): displayed in the language of structural forwardness, the energy landscape of the Mother Equation has a unique stable output manifold on the complex plane, namely the critical line.

Therefore, the three corroborate one another and together constitute the global structural coverage of the infinite zero set of the Riemann Hypothesis. The upper and middle volumes answer "why counterexamples cannot exist"; the lower volume answers "why the truth must be here". The two are the two faces of the same structural coin.

In summary, as a dual channel unitary system, the Mother Equation grows the critical line LA naturally on the complex plane through its geometric structure. Theorem 9.3A reduces the zero problem of doubly reflected entire functions to the zero distribution of the real entire function $g(w)$. The unitarity axiom of the Mother Equation, as the internal repulsion mechanism of the Whole 1, compresses the zero configurations from the complex plane onto the real axis, and then, through classical number theoretic facts, onto the negative half axis, which corresponds strictly to the critical line. The Riemann Hypothesis is not proved; it is grown, as the unique stable output of the self consistency of the doubly reflected symmetric structure. Upon the success of the physical contradiction self detonation and the success of the mathematical contradiction self detonation, the forward derivation grows successfully. Under the premise of pure growth from zero assumptions, the structured derivations and computations are correct, complete, and reproducible.

12.3 The Structural Rigidity of the Bidirectional Closure

The bidirectional closure is not a juxtaposition of a forward proof and a reverse proof; it is the two faces of the same structural coin. The syntactic structure of a traditional proof determines its unidirectionality:

$$\textit{premises} \Rightarrow \textit{rules} \Rightarrow \textit{conclusion}$$

Even with proof by contradiction, its logical form remains unidirectional:

$$\textit{assume a counterexample} \Rightarrow \textit{derive a contradiction} \Rightarrow$$

$$\textit{the counterexample does not exist} \Rightarrow \textit{the original proposition holds}$$

The "bidirectional proof" ($A \Leftrightarrow B$) in the traditional framework is in essence the juxtaposition of two independent unidirectional proofs: first prove $A \Rightarrow B$, then prove $B \Rightarrow A$, and finally place the two temporal derivation chains side by side. These two proofs are logically separable: what they share is not the same "now", but two distinct derivation sequences occurring one after another. The vulnerability of the conclusion of a "unidirectional proof" depends on the external subject's subjective acknowledgment of the premises (the truth of the premises, the legitimacy of the rules, the consistency of the meta system).

The reversible bidirectional closure under the framework of the Whole 1 of the Mother Equation is fundamentally different, in the epistemological dimension, from unidirectional proof:

Forward growth: $Whole\ 1 \rightarrow stable\ spectrum \rightarrow unique\ stable\ manifold = the\ critical\ line$

↑

↓

Reverse repulsion: $nonreal\ zeros \rightarrow \chi AB\ negative\ energy \rightarrow contradiction\ self\ detonation \rightarrow no\ living\ space$

The reverse repulsion (upper and middle volumes) is not a formalized proof by contradiction of "assume a counterexample exists and then derive a contradiction", but a direct determination: once a configuration of nonreal zeros is incorporated into the framework of the Mother Equation, it immediately forms an irreconcilable structural contradiction self detonation with the unitarity axiom at the definitional level. This configuration simply has no legitimate place of existence in the self consistency extension of the Whole 1.

The forward derivation (lower volume) is not a formalized deduction of "deriving the conclusion step by step from axioms", but a rigid attribute instantaneously exposed upon the completion of the structural definition of the Mother Equation: the critical line LA as the unique stable output manifold is not "discovered" but "grown".

The two are not two independent proofs "corroborating each other", but the same fact exposed simultaneously from

two inseparable directions by the self consistency extension of the Mother Equation as the Whole 1:

The reverse direction answers: why counterexamples cannot exist (nonreal zeros have no living space in the stable spectrum);

The forward direction answers: why the truth must be here (the critical line is the unique stable manifold satisfying unitarity).

The negative repulsion and the positive growth are not two coins each holding one face, but the two faces of the same structural coin. They share the definitional level of the Mother Equation, share the irreducible representations of the group algebra of V_4 , and share the representation theoretic invariant, the eigenvalue $-C$ on the χ_{AB} mode.

12.4 The Transcendence of the Bidirectional Closure over the Traditional Unidirectional Path to Rigor

The bidirectional closure of the Whole 1 of the Mother Equation locks the conclusion rigidly from both directions at once, forming the unescapability of the bidirectional closure:

Locking dimension	Reverse repulsion	Forward growth
Algebraic skeleton	energy of the χ_{AB} mode $-4(C - \varepsilon) < 0$ (a group algebra invariant)	$\{C + \varepsilon, \varepsilon\}$ both > 0 (strictly positive definite)
Group theoretic structure	four element orbit expanded, $\ \mathcal{O}(s)\ = 4, \text{Stab}(s) = \{\text{id}\}$	two element collapsed orbit, $\ \mathcal{O}(s)\ = 2, \text{Stab}(s) = \{\text{id}, \iota_A\}$
Geometric output	nonreal configurations have no legitimate place in the energy landscape	the critical line L_A is the unique stable output manifold

Should anyone raise a doubt, they must simultaneously deny:

- (1) the representation theoretic invariant that the eigenvalue of the group algebra of V_4 on the χ_{AB} representation is invariably $-C$;
- (2) the discrete jumps determined by the orbit stabilizer theorem ($|\text{Stab}| \in \{1, 2, 4\}$, no intermediate states);
- (3) the unviolability of the unitarity axiom as the definitional boundary of the very concept of a dual channel unitary system.

This triple locking is a rigidity fact at the level of the mathematical skeleton, not a disputable step at the level of derivation technique.

12.5 The Immediate Logical Fact: the Non Deferrability of the Bidirectional Closure

The truth of a traditional formalized proof is temporal, intersubjective, and deferrable. The conclusion requires the step by step audit of the expert community, the judgment of the meta system, and the verification of every link of the derivation chain.

The establishment of the Riemann Hypothesis exposed by the bidirectional closure of the Whole 1 of the Mother Equation, by contrast, is an immediate logical fact. It is not "a theorem derived within a system", but "a rigid attribute exposed simultaneously with the unfolding of the self consistency of the Whole 1". Its truth does not depend on the acknowledgment of the attitude of any external subject, does not depend on the extension of the temporal process, and does not depend on the choice of an axiom system. The reverse repulsion and the forward growth squeeze simultaneously from two directions until the unique possibility is rigidly locked: at the very moment when all

derivations and computations are verified or checked to be correct and the objective result is completed, the logical fact already holds; whether the conclusion holds can neither be deferred nor has any way to be deferred.

12.6 The Conclusion of the Main Text Derivation (the Upper, Middle, and Lower Volumes)

Under the constraint of the dual channel unitary structure of the Whole 1 of the Mother Equation, a configuration of nonreal zeros forms, at the definitional level, an irreconcilable structural contradiction with the unitarity axiom, and its living space is instantaneously excluded by the self consistency extension of the Whole 1; at the same time, the critical line $Re(s) = 1/2$, as the unique stable output manifold of the energy landscape of the Mother Equation, is strictly locked on the complex plane. Both directions have succeeded: the reverse repulsion and the forward growth are the two faces of the same structural coin, together constituting the global structural coverage of the infinite zero set of the Riemann Hypothesis. The Riemann ξ function naturally satisfies the sufficient conditions for entering the constraint spectrum of the Mother Equation, and its zero distribution therefore, as the natural unfolding of the structural self consistency of the Whole 1, lies strictly on the critical line. Therefore, when the physical contradiction self detonation of the upper volume succeeds, the mathematical contradiction self detonation of the middle volume succeeds, and the forward derivation of the lower volume succeeds, completing the unified closure of the three volumes, the reversible bidirectional closure is already formed, and the establishment of the Riemann Hypothesis is no longer "a theorem awaiting audit", but a logical fact instantaneously exposed with the unfolding of structural self consistency: the Riemann Hypothesis is instantaneously established.

Discussion

1. The Relation between Physical Self Detonation and Mathematical Self Detonation

Physical self detonation establishes the prerequisite mechanism of the intrinsic repulsion of the Mother Equation. As a dual channel unitary system, the unitarity axiom $\eta \equiv 1$ of the Mother Equation is not a theorem derived by inference, but the repulsion mechanism of the structure itself: the unitarity axiom $\eta \equiv 1$ requires the energy response to any nonzero perturbation to be strictly positive; while the geometric kernel on the four element orbit has eigenvalue invariably $-C < 0$ on the χ_{AB} representation (Part Four), causing configurations deviating from the critical line to produce negative energy $-4(C - \epsilon) < 0$, which forms an irreconcilable structural contradiction and conflict with the axiom at the definitional level, thereby self consistently rejecting all configurations deviating from the critical line. It must be specially pointed out that this self detonation closes completely within the Mother Equation and depends on no external mathematical object.

Mathematical self detonation translates the above constraint into the extension rule of doubly reflected entire functions. Theorem 9.3A reduces the zeros of doubly reflected entire functions to the real entire function $g(w)$; Lemmas 9.5A and 9.5C, through heat kernel regularization, strictly establish the lift from the delta kernel to the continuous gradient space, confirming that the eigenvalue on the χ_{AB} subspace is invariably $-C$; Theorem 9.5 shows that a nonreal zero necessarily activates this subspace, producing a strictly negative energy, and directly triggers the repulsion of the unitarity axiom of the Mother Equation. That is, the constraint rule already established by physical self detonation is imposed on all entire functions satisfying double reflection symmetry, confirming that their zero configurations fall into the forbidden spectrum of the Mother Equation.

The structural relation between physical self detonation and mathematical self detonation is that each is necessary and sufficient for the other. Physical self detonation provides the prerequisite repulsion mechanism, mathematical self detonation provides the concrete coverage: without physical self detonation, the mathematical derivation would stop at a neutral description ($-C$ being merely a negative number, not constituting a contradiction), that is, $-C$ would be merely a neutral operator spectrum without normative force; without mathematical self detonation, the constraint

of the Mother Equation would stop at abstract geometry and could not cover classical analytic objects. Therefore, physical self detonation provides the repulsion mechanism and mathematical self detonation provides the coverage; mechanism and coverage close at the definitional level, together constituting the global structural coverage of the infinite zero set. Should either one be missing, the structured reverse derivation of the Riemann Hypothesis cannot be completed.

2. The Epistemological Distinction between Counterexample Self Detonation Derivation and General Derivation, and the Immediate Logical Fact

The methodology adopted in this paper is not, in the traditional sense, a formalized proof, a proof by contradiction, or a general deductive derivation, but a counterexample self detonation derivation. Two fundamentally different logical operations must here be strictly distinguished. The validity of a general derivation depends on the external subject's subjective acknowledgment of the truth of the premises and the legitimacy of the rules of inference; its conclusion has transmitted force only within an intersubjectively agreed framework. Counterexample self detonation derivation, by contrast, cancels this subjective link entirely: an illegitimate configuration is not excluded through an external argument, but self detonates in contradiction at the structural definitional level with the axiom.

Concretely, the Mother Equation, as a dual channel unitary system, requires by its unitarity axiom that the energy response of the system to any nonzero local perturbation be strictly positive. Once a configuration of nonreal zeros is incorporated into the framework of the Mother Equation, it immediately generates an expanded four element orbit and activates the negative energy contribution of the corresponding representation mode; this energy contribution is strictly negative under the conceptual survival condition, directly forming a contradiction with the unitarity axiom. This repulsion mechanism is the direct logical consequence of the self definitional boundary of the dual channel unitary system; its necessity stems from the self existence conditions of the structure itself, and depends on no external hypothesis, no judgment of a meta system, and no accumulation of time.

Physical self detonation and mathematical self detonation together constitute the complete chain of the counterexample self detonation derivation. When a configuration of nonreal zeros triggers the definitional level contradiction with the unitarity axiom within the framework of the Mother Equation, the legitimate place of existence of that configuration in the structure is thereby eliminated. At this point, the absence of nonreal zeros is not a conclusion reached through the external subject's subjective acknowledgment, but the direct unfolding of the extension of the structural self consistency of the Mother Equation. The establishment of the Riemann Hypothesis is not a conclusion awaiting audit, but a logical fact instantaneously presented upon the completion of the counterexample self detonation derivation. Its truth does not depend on the acknowledgment of any subject's attitude or on the extension of any temporal process; it depends only on the completion of the operation of the double mechanism of physical self detonation and mathematical self detonation. This is the essence of the immediate logical fact: the conclusion holds at the very moment the derivation is completed, and can neither be deferred nor has any way to be deferred. Therefore, once the physical self detonation and the mathematical self detonation of the structured reverse derivation are established, the Riemann Hypothesis is instantaneously established.

In summary, the core theme of this paper is not to "find a nonreal zero and then prove that it does not exist", but to prove that "the configuration of nonreal zeros has no living space in the self consistency of the Mother Equation". Physical self detonation and mathematical self detonation do not "prove that the Riemann Hypothesis holds"; they "expose the contradiction self detonation of the counterexample in self consistency". Therefore, the Riemann Hypothesis is an immediately established logical fact, and not the result of a derivation or a mathematical proof.

Conclusion

(the five layer logical progression)

Layer one (the establishment of the logical fact):

Within the framework of structured reverse derivation, through the dual channel unitarity constraint of the Mother Equation (the Whole 1), this paper establishes the logical fact that the zeros of the Riemann zeta function can only lie on the critical line. This conclusion does not rely on zero by zero verification or finite approximation, but closes the existence space of the infinite zero set in one stroke through global structural coverage.

Layer two (the clarification of the essence of this logical fact):

The establishment of the above logical fact is not a valid derivation conducted within a bounded, optional axiom system, because the Mother Equation (the Whole 1) is not a set of optional formal axioms, but an indivisible structure defined as "logical self consistency itself". Its dual channel unitarity constraint is not a customized screening rule aimed at a specific object; it is in fact a repulsion mechanism (execution mechanism), and more precisely a structural unfolding of self consistency itself.

Layer three (the epistemological leap inferred from "the scope is infinite"):

Since the scope of the Mother Equation (the Whole 1) covers all doubly reflected symmetric entire functions and the entire space of their zero configurations, the conclusion "there are no nonreal zeros" forcibly produced by its self consistency no longer belongs to a local judgment of "holding within a system", but is a rigid attribute of this infinitely large logical extension itself: that is, the actual construction of this extension has never contained any living space for nonreal zeros.

Layer four (the positioning of the establishment of the Riemann Hypothesis as a fact):

The Riemann zeta function naturally lies within this logical extension. Therefore, the Riemann Hypothesis does not "hold within the framework set up in this paper"; rather, because the self consistency extension of the Whole 1 already contains the actual attribute of "no nonreal zeros", and because the Riemann zeta function happens to lie within this extension, the Riemann Hypothesis holds as the fact of an indivisible logical reality.

Layer five (the fundamental demarcation from formalized proof):

This conclusion is not "the validity of a theorem" in the sense of formalized proof, but "the exposure of a logical reality" in the sense of structured reverse derivation. It does not depend on the choice of an axiom system, does not depend on the stipulation of derivation rules, and does not depend on the judgment of a meta system; its sole ground is this: within the closed logical extension of definitional level self consistency, the counterexample has no living space. Therefore it is stronger and more thorough than the validity of any formalized proof, because what it reveals is not a derivable proposition within some system, but the construction boundary of reality itself.

That is, the actual holding of the logical fact has validity higher than and greater than the establishment of a mathematical proof. Physical self detonation is the internal tension of the self consistent structure; mathematical self detonation is the rigid conflict between the zero configurations and the structural skeleton; the Riemann zeta function (the ζ function) naturally lies within the extension of this structure because of double reflection symmetry; the nonreal zeros have no living space in this extension. In summary, within the framework of structured reverse derivation, through the dual channel unitarity constraint of the Mother Equation (the Whole 1), this paper establishes the logical fact that the zeros of the Riemann zeta function can only lie on the critical line. This conclusion does not

rely on zero by zero verification or finite approximation, but closes the existence space of the infinite zero set in one stroke through global structural coverage. The Riemann Hypothesis does not hold within some framework; rather, because the logical extension of the self consistency of the defined "Whole 1" itself already contains the fact of "no nonreal zeros", and because the Riemann zeta function lies precisely within this extension, the Riemann Hypothesis is instantaneously established as a fact. Therefore, the Riemann Hypothesis holds as a fact, as a rigid attribute of this logical reality. The Riemann Hypothesis is instantaneously established in the reality of the logical fact.

Finally, it is particularly worth noting that, as a dual-channel unitary system, the Mother Equation requires only entireness, strict central symmetry without multipliers, and Schwarz reflection as entry conditions for its constraint spectrum, and is not specific to the Riemann ξ function. This means that any doubly reflected entire function satisfying the above three conditions, including the ξ function corresponding to the original Riemann Hypothesis and certain completed forms of self-dual L-functions (such as real primitive Dirichlet L-functions and real Hecke L-functions), naturally falls into the constraint spectrum of the Mother Equation, and its zero distribution is subject to the same unitarity mechanism. However, to implement the above general structure as a complete proof or derivation for specific L-functions, independent arithmetic analyses (such as real-axis positivity verification) remain necessary for each object; although the present work has laid a considerable foundation for this purpose, this lies beyond the scope of the present paper, which focuses on the original Riemann Hypothesis, and will be built upon in future work.

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